

MIMO over Satellite: A Review

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Abstract—The present article carries out a review of MIMO-based techniques that have been recently proposed for satellite communications. Due to the plethora of MIMO interpretations in terrestrial systems and the particularities of satellite communications, this review is built on two pillars, namely fixed satellite and mobile satellite. Special attention is given to the characteristics of the satellite channel, which will ultimately determine the viability of MIMO over satellite. Finally, some future research directions are identified.

Index Terms—MIMO, single-user, multi-user, space-time codes, precoding, satellite channel modeling.

I. INTRODUCTION

THE SUCCESS story of MIMO technology in terrestrial applications now spans more than a decade with spectacular results, as it offers substantial leverage in realizing the next generation gigabit wireless [1]. The reason for the respective intensive research is that MIMO technology offers many advantages and degrees-of-freedom, such as: (a) space and multiuser diversity gain, (b) spatial multiplexing gain, (c) array and coding gain, and (d) interference reduction. The salient feature of MIMO-based systems is that the above merits come from an information theory point of view at no extra cost concerning transmit power or bandwidth. Instead, MIMO-based systems take advantage of what is probably the last unexploited frontier in wireless communications, the *spatial domain*. A testimony to its success is that different aspects of MIMO technology are being planned or have already been incorporated in wireless terrestrial standards, such as the IEEE 802.11n, 802.16e, 802.16m, 802.20, 802.22, 3GPP Releases 7, 8 (LTE) and 99, 3GPP2 UMB, DVB-T2 among others.

In an effort to remain competitive with terrestrial systems, SatCom are trying to follow the progress in terrestrial MIMO technology and profit from the significant research achievements in the area of multiple antenna techniques. However, MIMO is a rather generic term that encompasses a plethora of techniques including broad categories such as *single-user* (SU), *multi-user* (MU) and *distributed/virtual* MIMO. Hence,

the question to answer is what particular MIMO technique is applicable to SatCom, since the latter exhibit distinct characteristics compared to terrestrial systems, with regard to service coverage, link geometry, propagation delay, channel impairments, interference scenarios and physical layer interface. Moreover, we can distinguish between different SatCom systems variants depending on [2], [3]: the choice of orbit (GSO vs. NGSO), user mobility (fixed vs. mobile), operating frequency band (UHF, L, S, C, X, Ku, Ka bands), group size of intended users (broadcast, multicast, unicast), multiplexing scheme (single carrier TDM vs. multicarrier OFDM), type of application (delay tolerant vs. delay intolerant), availability of FMTs (CCM vs. ACM) and so on.

This ambiguous landscape regarding the applicability of MIMO over satellite has motivated the present review article in an attempt to provide a thorough comparative classification of already proposed techniques as well as possible future research perspectives. The review is built around two characteristic cases of satellite systems, which are nowadays driving the commercial development of SatCom:

- 1) *Fixed satellite* (FS) systems operating over GSO orbits at frequency bands above 10 GHz (e.g. Ku, Ka) serving *fixed satellite terminals* (FSTs) in an unobstructed propagation environment.
- 2) *Mobile satellite* (MS) systems operating over GSO orbits at frequency bands well below 10 GHz (e.g. L, S) serving *mobile satellite terminals* (MSTs) in propagation environments suffering from different degrees of obstruction (urban, suburban, rural).

The above two broad cases constitute the main fields of application of the very successful satellite standards recently developed by ETSI, namely the DVB-S2 standard [4], [5] for FS systems and the DVB-SH standard [6], [7] for MS systems. Also, similar study cases have been selected by ESA to investigate the application of MIMO over satellite in a series of recent technical studies [8], [9], [10].

The review article is organized as follows: Section II provides a synopsis of MIMO in terrestrial systems. It is *not* intended to carry out a survey of MIMO—there are numerous excellent textbooks (e.g. [11], [12]) and survey articles (e.g. [1], [13], [14], [15]) that serve this purpose—but to provide an easy reference to the various MIMO techniques when discussing their application in the satellite domain; this constitutes the main contribution of this article. As the performance of any MIMO technique depends drastically on the underlying channel characteristics, Section III describes the dominant propagation characteristics influencing FS and MS systems, putting emphasis on the state-of-the-art

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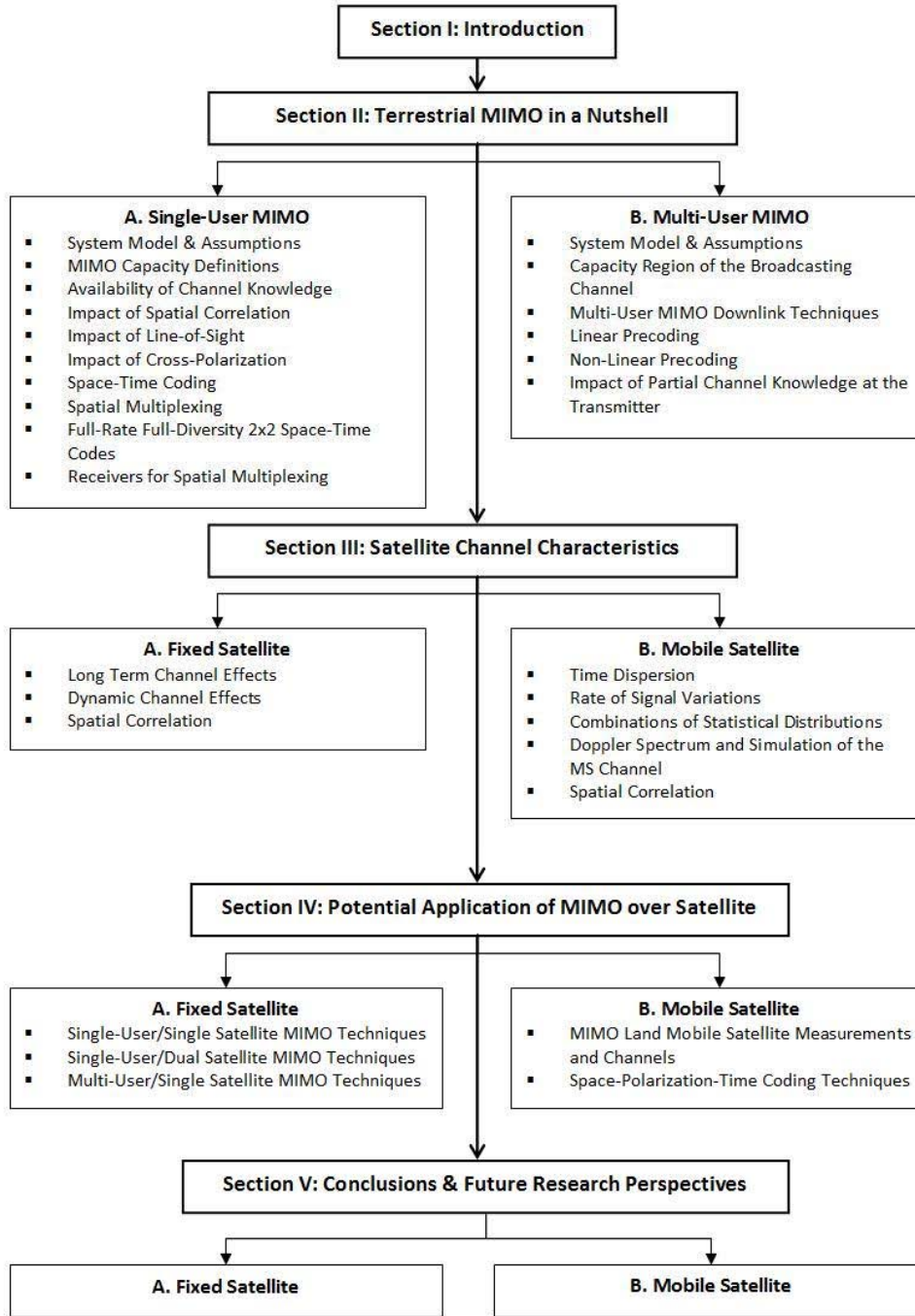


Fig. 1. Organization of the review article.

in satellite channel modeling. Note that the satellite channel characteristics, apart from substantially differing from those of terrestrial systems, are also very different when FS and MS systems are considered. Section IV presents the various research approaches published in the literature on how to apply MIMO over satellite, distinguishing again between FS and MS. Finally, Section V concludes the article and presents some thoughts about future research on still underexploited aspects of MIMO over satellite. The overall organization of the article including all individual topics discussed per subsection is presented in the block diagram of Fig. 1.

A. Mathematical Notation

For the purposes of this article, a number of notations are necessary. Vectors are written in boldface lower case letters; matrices in boldface capital letters. Superscripts T , $*$, H and † denote transposition, elementwise conjugation, conjugate transposition and the Moore-Penrose pseudoinverse of a matrix, respectively. $E[\cdot]$ denotes the expectation operator and $*$ denotes the convolution operator. $\mathbf{I}_m$ stands for the $m \times m$ identity matrix, $\mathbf{0}$ denotes the all zeros matrix of appropriate dimensions. $\|\mathbf{A}\|_F$, $\det(\mathbf{A})$, $\text{rank}(\mathbf{A})$, $\text{diag}(\mathbf{A})$ and $\text{Tr}(\mathbf{A})$

stand for the Frobenius norm, determinant, rank, diagonal and trace of matrix $\mathbf{A}$, respectively. $\|\mathbf{a}\|$ denotes the Euclidean norm of vector $\mathbf{a}$. For a $m \times n$ matrix $\mathbf{A} = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$, the vector $\text{vec}(\mathbf{A}) = [\mathbf{a}_1^T \ \mathbf{a}_2^T \ \cdots \ \mathbf{a}_n^T]^T$ stacks $\mathbf{A}$ into a vector columnwise. The Kronecker product of two matrices is denoted by $\otimes$, while $\odot$ is used to represent the Hadamard product of two matrices.

II. TERRESTRIAL MIMO IN A NUTSHELL

This introductory section outlines the various MIMO techniques studied for terrestrial systems that have also been proposed for possible application over satellite. Due to its limited scope, it does *not* provide a general survey on MIMO, but will be used as a reference to analyze the MIMO techniques specifically proposed for application over satellite in Section IV.

A. Single-User MIMO

SU-MIMO System Model & Assumptions: Multiple antenna techniques in point-to-point configurations are commonly referred to as SU-MIMO. Consider a general MIMO system with M_T transmit and M_R receive antennas (in short $M_R \times M_T$ MIMO) as shown in Fig. 2. The time-varying channel matrix for every combination of receive and transmit antenna is denoted by $\mathbf{H}$. If for convenience the symbol period T_S is set equal to 1 s, the discrete time input-output relation in a frequency flat (narrowband) MIMO channel¹ is:

$$\mathbf{y}[k] = \sqrt{\frac{E_S}{M_T}} \mathbf{H} \mathbf{s}[k] + \mathbf{n}[k], \quad (1)$$

where the index k corresponds to the k th symbol period (burst)² and E_S is the average transmit symbol energy³. $\mathbf{s}(k) = [s_1(k) \ \cdots \ s_{M_T}(k)]^T$ is the $M_T \times 1$ vector of signals transmitted from each antenna, $\mathbf{y}(k) = [y_1(k) \ \cdots \ y_{M_R}(k)]^T$ is the $M_R \times 1$ vector of signals received by each antenna and $\mathbf{n}(k)$ is the $M_R \times 1$ noise vector of independent ZMCSCG random variables with variance N_0 . Downsized versions of a MIMO system with a single transmit antenna, a single receive antenna or single antennas at both ends are termed SIMO, MISO or SISO, respectively.

MIMO Capacity Definitions: Earlier works on the capacity of SU-MIMO from an information theory perspective [16], [17] demonstrated that tremendous capacity gains may result from ideal rich scattering environments and motivated extensive research over the next years. These initial pioneering works resulted in the well-known "log-det" capacity formula, which for a deterministic MIMO channel characterized by a channel matrix $\mathbf{H}$ of dimension $M_R \times M_T$ is written as:

$$C = \max_{\mathbf{Q}} \left\{ \log_2 \det \left(\mathbf{I}_{M_R} + \frac{E_S}{M_T N_0} \mathbf{H} \mathbf{Q} \mathbf{H}^H \right) \right\} \quad (2)$$

In (2), the covariance matrix of the transmitted signal vector $\mathbf{s}$, $\mathbf{Q} = E[\mathbf{s}\mathbf{s}^H]$, satisfies the constraint $\text{Tr}(\mathbf{Q}) = M_T$

¹In general, this article focuses on narrowband channels, as is usually the case for SatCom.

²For simplicity and clarity, in the rest of the analysis the time index k is dropped.

³Given that $T_S = 1$ s, E_S is also the average transmit power.

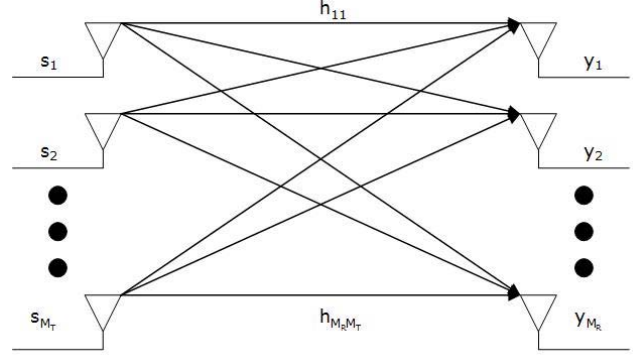


Fig. 2. MIMO system employing M_T transmit and M_R receive antennas.

concerning the total average energy transmitted over a symbol period. In fading channels, the complex channel gain h_{ij} included in $\mathbf{H}$ vary with time. In this case, two relevant definitions of capacity exist, namely the *ergodic capacity* and the *outage capacity*, which apply to fast and slow fading channels, respectively. The ergodic or Shannon capacity [18] defines the maximum rate, -averaged over all channel realizations-, which can be transmitted over the channel based only on the distribution of $\mathbf{H}$. In other words, it corresponds to the expected value of the capacity in (2) assuming that power is optimally allocated. With regard to the outage capacity, the transmitter fixes a transmission rate R and the outage probability associated with R is the probability that the transmitted data will not be received correctly, that is $P_{out} = \Pr\{\mathbf{H} : C < R\}$.

Availability of Channel Knowledge: The basic capacity formula (2) depends on the degree of channel knowledge or CSI at the transmitter (CSIT) and the receiver (CSIR) [19]. The degree of CSI varies from no CSI up to full (or perfect) CSI depending on whether exact channel gain values are available at the transmitter/receiver for every channel realization or only a statistical measure of the channel is available (CDI). In the exceptional case where the channel is perfectly known at both link ends, the MIMO channel may be decomposed into $\text{rank}(\mathbf{H})$ parallel SISO channels obtained through SVD. The transmitter can access these spatial subchannels through the optimal energy allocation applying the so-called *waterfilling algorithm* [20]. Anyway, perfect CSIR is usually assumed in MIMO systems since the channel gains can be estimated fairly easily through pilot sequences. In this case, the strategy that maximizes capacity is to allocate equal power to each transmit antenna [16], that is $\mathbf{Q} = \mathbf{I}_{M_T}$. In fading channels with a ZMCSCG distribution, the ergodic capacity reveals that, albeit no CSIT exists, the MIMO channel capacity grows (approximately) linearly with $\min(M_T, M_R)$. This remarkable outcome is the main reason for the popularity of MIMO techniques, as they achieve high data rates at no extra transmit power or bandwidth. At the absence of CSI at both the transmitter and the receiver, the linear capacity increase with $\min(M_T, M_R)$ disappears and, in some cases, adding more antennas offers a negligible capacity gain [21]. An in between situation with

respect to no and full CSIT knowledge is the exploitation of CDI instead of the instantaneous CSI at either the transmitter (CDIT) or the receiver (CDIR); the capacity under different combinations of CDI has been documented in detail in the literature [22], [23], [24], [25]. CDI provides a practical solution when reliable channel estimation is not feasible, since the channel statistics change slower than the channel itself, rendering this information easier to obtain.

Impact of Spatial Correlation: The IID ZMCSCG channel, usually denoted by $\mathbf{H}_w$, results from a rich scattering environment with sufficient antenna spacing at the transmitter and receiver. In practice, however, the $\mathbf{H}_w$ assumption is seldom true and some degree of spatial correlation exists. The so-called Kronecker model [26] assumes that the spatial correlations at the transmitter and receiver are distinguishable:

$$\mathbf{R} = \mathbf{R}_T \otimes \mathbf{R}_R, \quad (3)$$

where $\mathbf{R}_T = E[\mathbf{H}^H \mathbf{H}]$ is the $M_T \times M_T$ transmit covariance matrix, $\mathbf{R}_R = E[\mathbf{H} \mathbf{H}^H]$ is the $M_R \times M_R$ receive covariance matrix and $\mathbf{R} = E[\text{vec}(\mathbf{H}) \text{vec}(\mathbf{H})^H]$ is the $M_T M_R \times M_T M_R$ total covariance matrix. Hence, the general form of matrix $\mathbf{H}$ is:

$$\mathbf{H} = \mathbf{R}_T^{1/2} \mathbf{H}_w \mathbf{R}_R^{1/2} \quad (4)$$

High spatial correlation is detrimental to MIMO capacity [27].

Impact of Line-of-Sight: LOS or, equivalently, Ricean fading is conveniently included in the MIMO matrix through expressing $\mathbf{H}$ as the sum of a purely deterministic (fixed/LOS) component/matrix and a zero-mean stochastic (variable/NLOS) component/matrix:

$$\mathbf{H} = \sqrt{\frac{K}{K+1}} \bar{\mathbf{H}} + \sqrt{\frac{1}{K+1}} \mathbf{H}_w, \quad (5)$$

where K is the Rice factor. In terms of capacity at high values of the K factor, if the fixed component of the channel is rank deficient, capacity increases only logarithmically with $\min(M_T, M_R)$. However, as suggested in [28], a linear capacity growth is still achievable in a LOS environment under a specific geometrical arrangement of the transmit elements.

Impact of Cross-Polarization: Based on the concept of polarization diversity in mobile radio [29], the use of antennas with multiple polarizations can overcome possible space limitations due to multiple antennas at the BS and at the mobile terminal and still achieve the advantages predicted by MIMO theory. Note that antenna spacings of ten wavelengths and at least half a wavelength are necessary at the base and the mobile stations, respectively. For a 2×2 MIMO system with two orthogonal polarizations (either $0^\circ/90^\circ$ or $\pm 45^\circ$), the diagonal elements of $\mathbf{H}$ correspond to transmission and reception on the same polarization, while the off-diagonal elements correspond to transmission and reception on orthogonal polarizations. The power of the individual channel elements is [30]:

$$E[|h_{11}|^2] = E[|h_{22}|^2] = 1 \quad (6)$$

$$E[|h_{12}|^2] = E[|h_{21}|^2] = \alpha, \quad (7)$$

where α ($0 \leq \alpha \leq 1$) depends on the XPD⁴ due to both the antenna design and the propagation environment. Assuming Rayleigh fading and taking into account (4), $\mathbf{H}$ with cross-polarized antennas can be approximately modeled by [11]:

$$\mathbf{H} = \mathbf{X} \odot \mathbf{R}_R^{1/2} \mathbf{H}_w \mathbf{R}_T^{1/2}, \quad (8)$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & \sqrt{\alpha} \\ \sqrt{\alpha} & 1 \end{bmatrix} \quad (9)$$

In general, the concept of using polarization to provide two or even more distinct communication channels is one of the milestones for generating MIMO over satellite (see Section IV). Early works suggested that any of the six electric and magnetic field vector polarizations may offer an independent channel [31].

Space-Time Coding: The information theoretic capacity analysis does not reflect the actual performance achieved by real transmission systems, since it only provides upper bounds foreseen by algorithms/codes with boundless complexity or latency. In practice, codes with a reasonable compromise between error rate and complexity are required to realize MIMO gains. These codes are known as STC and aim at improving either the link *reliability* by providing diversity gain or the *throughput* by providing multiplexing gain. What is most important is that these gains are achieved having available CSIR only and no CSIT.

The discussion about STC starts with STBC, particularly with the basic form of these codes, the *Alamouti scheme* [32], a simple but ingenious transmit diversity technique that does not require CSIT. According to the Alamouti scheme, for a 1×2 MISO geometry, two different symbols s_1 and s_2 are simultaneously transmitted from antennas 1 and 2, respectively, during the first symbol period, followed by symbols $-s_2^*$ and s_1^* from antennas 1 and 2 during the next symbol period. Assuming that the channel remains constant over two symbol periods, it turns out that diversity of order 2 (full diversity) is extracted, even in the absence of CSIT. However, this scheme does not provide any array gain. The Alamouti scheme can be generalized to MIMO systems with up to four antennas using orthogonal codewords, namely OSTBC [33]. STTC introduced in [34] are an extension of the conventional trellis codes to multiantenna systems. Unlike STBC, STTC can achieve both full spatial diversity and coding gain [35]. Each STTC can be described using a trellis diagram, with the number of nodes corresponding to the number of states in the trellis. To decode STTCs, a multidimensional Viterbi algorithm is employed at the receiver, which renders decoding highly complex.

Spatial Multiplexing: When STC are intended for diversity, one or less independent symbols is transmitted per symbol period. On the other hand, when codes are designed for *spatial multiplexing*, M_T independent symbols per symbol

⁴Poor XPD results in a value of α close to 1 whereas high XPD results in $\alpha \rightarrow 0$.

period are transmitted. This is usually carried out employing one of the variants of BLAST coding that follow:

- *Diagonal encoding* (D-BLAST) [36], where the data stream is first serial-to-parallel demultiplexed onto M_T separate streams. Each stream undergoes independent temporal coding, interleaving and symbol mapping and, then, it is fed into a stream rotator that rotates the symbols diagonally across antennas and time.
- *Parallel encoding* (V-BLAST) [37] is a simplified version of D-BLAST aiming at a lower computational complexity by removing the stream rotator.
- *Serial encoding* is another variant of D-BLAST, where the input bit stream is first SISO encoded, interleaved and mapped to a constellation point and then serial-to-parallel demultiplexed onto the M_T antennas.
- *Turbo-BLAST* [38], [39] combines layered STC with the Turbo coding principle. It overcomes the limitation of V-BLAST of having more transmit than receive antennas and it is based on a random layered STC scheme and a Turbo-like decoder of the random layered STC.

Full-Rate Full-Diversity 2x2 Space-Time Codes: Practical reasons limit current wireless communication systems to small dimensions, typically 2×2 . In this regard, various 2×2 STC have been proposed in the literature that, in contrast to Alamouti and BLAST, obtain the full-rate, full-diversity frontier. The most characteristic is the 2×2 Golden STBC code introduced in [40], although also studied independently in [41], [42]. Golden codes, based on cyclic division algebra, possess many desired properties: full rate (2 symbols per channel use), full diversity (equal to 4), non-vanishing minimum determinant independent of the constellation size and preserve the spectral efficiency. In fact, the IEEE 802.16e-2005 specification includes a variant of the Golden codes dubbed as Matrix C, since Matrix A and B correspond to Alamouti and spatial multiplexing. The main drawback of Golden codes is the high decoding complexity, which grows with the fourth power of the signal constellation size.

Receivers for Spatial Multiplexing: In the analysis of MIMO techniques over satellite various kinds of receivers appear. To reduce the complexity of the optimal (full search-based) ML receivers, researchers have resorted to *linear techniques*, such as ZF and MMSE. Furthermore, the key idea of the *SIC decoder* is layer peeling, i.e. symbol streams are successively decoded and stripped away layer by layer. This strategy, called *interference nulling and canceling*, is detailed in [36], [43], [44] and gives a reasonable tradeoff between complexity and performance. *Iterative receivers* may approach optimum performance at an affordable receiver complexity. Iterative receivers for MIMO are based on a combination of iterative interference cancelation and coding. Finally, *sphere decoding* aims at reducing the computational complexity of ML by narrowing down the size of the lattice searched. Though the worst case complexity remains exponential with respect to the constellation size (as in ML), the expected complexity of sphere decoding has been shown to be cubic or even sub-cubic at high SNR [45].

B. Multi-User MIMO

MU-MIMO System Model & Assumptions: Multiple antenna techniques in point-to-multipoint systems are commonly referred to as a MU-MIMO. MU-MIMO exhibits key advantages over SU-MIMO, such as [14]:

- In addition to stream multiplexing, MU-MIMO schemes offer MU multiplexing, resulting in a direct capacity gain proportional to the number of BS antennas and the number of users.
- MU-MIMO appears more immune to most of the adverse propagation phenomena resulting in a low rank SU-MIMO channel matrix, such as LOS or antenna correlation. Although increased correlation affects the diversity achieved by each user, MU diversity can be extracted instead.
- MU-MIMO allows for spatial multiplexing gain at the BS without necessitating terminals with multiple antennas. This is especially significant from a commercial point of view, since cost is kept in the infrastructure side.

The above advantages of MU-MIMO come at a certain cost:

- Perhaps the most important drawback is that MU-MIMO requires CSIT to perform spatial multiplexing. While not essential in SU-MIMO, CSIT is of utmost importance in this case.
- The users in MU systems may significantly differ with respect to the channel conditions. This gives rise to fairness issues related to the selection of the subgroup of users that will be served –*scheduling*.
- In SU-MIMO, coding at the transmitter and decoding at the receiver can be done in a cooperative fashion, since the respective multiple antennas are co-located, whereas in MU-MIMO users are geographically dispersed.

The MU channels can be distinguished into: the *broadcasting channel* (BC), which is the downlink from the BS to the mobile terminals, and the *multiple access channel* (MAC), which is the corresponding uplink (see Fig. 3). The rest of this article will focus on the BC. Consider a system with M antennas at the BS and K users, each equipped with N_k antennas, $k = 1, \dots, K$. Assuming frequency flat fading, the downlink (BC) channel from the BS to the k th user is represented by a complex Gaussian $N_k \times M$ matrix $\mathbf{H}_k$. The $N_k \times 1$ signal vector received at the k th user can be written as:

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{s} + \mathbf{n}_k, \quad (10)$$

where $\mathbf{s}$ represents the $M \times 1$ signal vector transmitted from the BS and $\mathbf{n}_k$ is the $N_k \times 1$ ZMCSCG additive noise at receiver k . The transmit covariance matrix of the input signal is $\mathbf{Q} = E[\mathbf{s}\mathbf{s}^H]$ and the BS is subject to an average power constraint $\bar{P}$, which implies $\text{Tr}(\mathbf{Q}) \leq P$.

Capacity Region of the Broadcasting Channel: The region of all user rates that are simultaneously achievable is the *capacity region* [46]. A rate vector $[R_1, \dots, R_K]$ is *achievable* if a coding scheme exists assuring that the error probability of all users goes to zero as the code block length becomes long enough. An important point on the boundary of the capacity region is the *sum rate point*, which corresponds

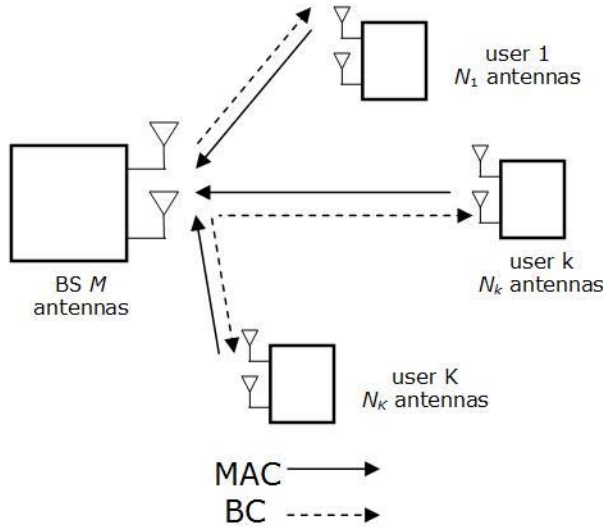


Fig. 3. The MU-MIMO MAC and BC.

to the maximum of the rates that can be conveyed by the transmitter to the receivers, often called the *throughput* of the system.

To determine the capacity region for a general BC has been quite challenging, since the MIMO BC with Gaussian noise falls under the category of nondegraded BC, the capacity of which is generally unknown. Therefore, progress in this area [47], [48], [49], [50] has been one of the major achievements in information theory over the last years. The difficulty in assessing the capacity region of the BC stems from the fact that, although the interference at each receiver due to signals intended for other receivers is known at the transmitter (since it is the transmitter that generates all these signals) and, hence, the transmitter can potentially pre-subtract all interference, this may not be done without violating the power constraint. In a surprising result [51], it was shown that, when noise and interference are Gaussian, an interference pre-subtraction scheme called *dirty paper coding* (DPC) achieves the same capacity as if interference is not present.

The asymptotic analysis carried out in [52] reveals various scaling laws for the BC sum rate capacity: Under full CSIT, the system capacity in the high SNR region increases linearly with the number of transmit antennas at the BS, provided that $K > M$, which is a reasonable assumption. Moreover, keeping the number of transmit antennas M and the power constraint P constant and assuming full CSIT and CSIR, it has been shown [53] that:

$$\lim_{K \rightarrow \infty} \frac{C_{BC}}{\log \log K} = M \quad (11)$$

The above total multiplexing gain $M \log \log K$ (double-logarithmic with respect to the number of users) obtained for a large number of users is attributed to the inherent *MU diversity* of the system. Another interesting finding from the satellite application point of view is that, in contrast to SU-MIMO, the number of receive antennas plays a minor role in the sum rate capacity of the downlink. Consequently,

as far as the sum rate is concerned, adding receive antennas to the users is not beneficial.

Multi-User MIMO Downlink Techniques: In MU-MIMO systems, the physical separation of the receivers in the BC prevents joint receive processing; consequently, joint suppression of MU interference must be performed at the transmitter. This means that, in contrast to the MIMO MAC, to optimize MIMO BC, CSIT must be available at the BS. Although DPC may achieve the sum rate capacity of the MIMO BC, its implementation is difficult in real systems. Therefore, practical techniques have been developed to process the MU-MIMO downlink signal at the BS. These suboptimal implementations of DPC are collectively termed as *downlink precoding* or *beamforming* techniques. The basic idea behind this approach is that MU-MIMO downlink interference due to signals transmitted to other users is known at the transmitter and, in principle, a precoder can be used to annihilate this effect. The description of BC precoding techniques is of increased interest, since the forward link of a satellite system⁵ can be modeled as a MIMO BC.

Linear Precoding: Linear precoding in MU systems can be viewed as a generalization of the traditional SDMA, where users are assigned different precoding matrices at the transmitter. Precoders are designed jointly based on the CSI of all the users employing different criteria, such as error probability, SINR, sum rate capacity etc. Based on the knowledge of the channel matrix, the goal of linear precoding is to design the precoding matrix $\mathbf{F}$ aiming either at maximizing the sum of the information rates for all users subject to a sum transmit power constraint or at minimizing the total transmitted power while achieving a specified QoS.

Non-Linear Precoding: Linear precoding based on plain channel inversion performs poorly for large numbers of users and receive antennas, thus remaining far from the optimal DPC sum capacity predicted by information theory. Nonlinear precoding involves additional transmit signal processing to improve error rate performance. To support the analysis to be presented in Section IV, it is worthwhile to discuss a representative non-linear technique, namely the THP technique [54]. Originally introduced for ISI pre-equalization in SISO channels, THP can be readily extended to MU-MIMO [54] and can be interpreted as moving the feedback part of the typical DFE to the transmitter. The THP architecture consists of a forward filter $\mathbf{F}$, a backward filter $\mathbf{B}$, a modulo operator and a diagonal weighted filter $\mathbf{G}$ (see Fig. 4)⁶. Unlike linear precoding methods, THP can be applied in cases when the aggregate number of receive antennas is not less than the number of transmit antennas, i.e. $NK \geq M$.

⁵The forward link of a SatCom system comprises the uplink between the GW station and the satellite and the downlink between the satellite and the FSTs or MSTs.

⁶Depending on the position of $\mathbf{G}$, Fig. 4 illustrates the two basic structures—one is to place the entries of $\mathbf{G}$ at the receivers in a decentralized manner (*decentralized THP*) and the other is to place $\mathbf{G}$ at the transmitter in a centralized manner (*centralized THP*)—.

Impact of Partial Channel Knowledge at the Transmitter: As already emphasized, having CSIT available is imperative to obtain full benefit from MU-MIMO. In an effort to reduce the amount of feedback required at the transmitter (partial channel knowledge) and still grasp many of the advantages of MU-MIMO, *limited feedback communications* [55] have produced significant results [56]. In general, most of these recent techniques involve some kind of feedback quantization or are enhanced MU versions of *opportunistic beamforming* [57]. Random opportunistic beamforming offers a way of handling both the beamforming and scheduling problems simultaneously with scalar only SNR feedback. In this scheme [53], once M orthonormal beams are generated at random, each user calculates its SINR for each of the M beams and feeds back its best SINR value along with the corresponding beam index. The best user of each beam is then scheduled.

III. SATELLITE CHANNEL CHARACTERISTICS

Channel and propagation characteristics are the major constituents of a MIMO channel matrix, which crucially determine the performance of any potentially adopted MIMO technique. Along these lines, a review of the most important spatial and temporal channel characteristics affecting propagation for FS and MS is imperative.

A. Fixed Satellite

FS communication systems above 10 GHz operate under LOS; the satellite channel essentially corresponds to an AWGN channel. However, on top of this, propagation at the Ku and, especially, Ka band is subjected to various atmospheric fading mechanisms originating in the troposphere, which severely degrade system performance and availability. These adverse tropospheric phenomena, well documented among others in [58], [59], [60], are briefly summarized in the following, distinguishing between *long term* and *dynamic channel effects*. Emphasis is placed on state-of-the-art propagation models developed primarily within Study Group 3 of the ITU-R, which is responsible for radiowave propagation.

Long Term Channel Effects: The most important channel effects impairing SatCom at frequencies above 10 GHz are summarized as follows [60]:

- *Attenuation due to precipitation:* When propagating through snow, hail, ice droplets and, predominantly, rain, radiowaves suffer from hydrometeor scattering and absorption. This results in a flat and slow fading process proportional in dB to the square of the frequency. Hence, rain attenuation constitutes the dominant factor limiting FS system availability. Its effect can be predicted employing the empirical model proposed in ITU-R Recommendation P.618 [61].
- *Gaseous absorption:* Absorption from oxygen and water vapor contributes to the total attenuation, though to a much smaller extent than rain attenuation. A complete method for the calculation of this impairment is given in Annex 1 of ITU-R Recommendation P.676 [62].

- *Cloud attenuation:* The liquid water content of clouds is the physical cause of this type of attenuation. Prediction models for this particular attenuation factor have been developed within the framework of ITU-R Recommendation P.840 [63].
- *Tropospheric scintillations:* This fast fading mechanism due to variations in the refractive index of the troposphere is aggravated as the frequency of operation increases. An empirical model estimating the effect of scintillations on the received signal can be found in [61].
- *Signal depolarization:* Differential phase shift and attenuation caused by nonspherical scatterers such as rain drops and ice crystals result in significant depolarization [64]. As a result, part of the transmitted power in one polarization interferes with its orthogonal counterpart and the resulting XPD can be predicted employing the empirical model proposed in ITU-R Recommendation P.618 [61].
- *Sky Noise Increase:* As attenuation increases, so does emission noise. The same factors previously mentioned, i.e. scatter/emission from precipitation hydrometeors, contribute to noise increase.
- *Total attenuation:* The performance degradation due to the above phenomena, necessitates especially in the Ka band the use of a total prediction model that takes into account the combined effect of the various attenuation factors taking into account how the individual propagation effects depend on each other. A combination method that considers some of the effects as being uncorrelated is proposed in [61]. Based on this, the distribution of the total attenuation over the satellite link is given by:

$$A_{tot}(p) = A_g(p) + \sqrt{[A_c(p) + A_r(p)]^2 + A_s^2(p)}, \quad (12)$$

where $A_g(p)$, $A_c(p)$, $A_r(p)$, and $A_s(p)$ represent gaseous, cloud, rain and scintillation attenuation for $p\%$ of annual time, respectively. Fig. 5 presents an application of the rain and total attenuation models at the Ka band according to [61] for the equatorial area of Singapore, where tropospheric phenomena are particularly pronounced.

Dynamic Channel Effects: Apart from the previous empirical distributions predicting the long term behavior of the satellite channel (*first order statistics*), significant research efforts have been addressed toward developing stochastic models that accurately reproduce the temporal properties of the AWGN channel when impaired by rain fading. Dynamic channel models allow for the calculation of several *second order statistics*, such as fade slope and fade duration. In turn, this type of stochastic modeling offers a powerful tool for the simulation of adaptive FMTs [60], [65]. Various models have been proposed for the generation of rain attenuation time series [66]. However, there is a general consensus that a good statistical agreement with measured data is achieved by employing the Maseng-Bakken stochastic model [67], updated in [68], [69]. A snapshot

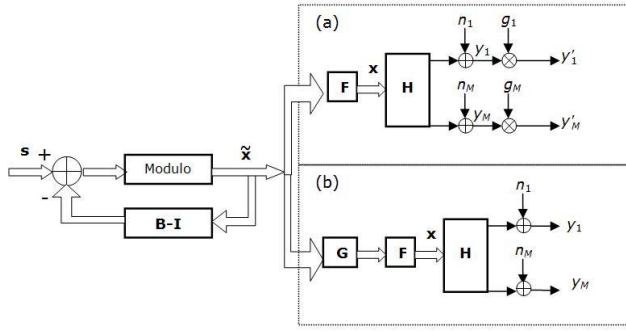


Fig. 4. Block diagram of the MU-MIMO downlink with THP. (a) Decentralized structure: The diagonal weighted matrix $\mathbf{G}$ is separately placed at the receiver and each receiver has a diagonal entry g_k . (b) Centralized structure: The diagonal weighted matrix $\mathbf{G}$ is placed at the transmitter.

of a typical rain event from a rain attenuation time series synthesizer based on these assumptions is illustrated in Fig. 6.

Spatial Correlation: The essence of MIMO technology is the utilization of the spatial dimension to improve the performance of wireless systems. Actually, as discussed in Section II, the predicted performance gains achieved by MIMO are fully exploited when the amount of correlation between the alternative paths formed by the multiple antennas at the transmitter and the receiver is low. The spatial structure of rain exhibits inhomogeneity, a property that has been exploited since the 1970s by SD and OD (or SatD⁷) [70], [71] to improve the quality of reception. However, the fact that Earth-space links operating above 10 GHz are LOS, i.e. there are no nearby scatterers at the satellite or the earth station to cause multipath, makes it difficult to achieve fully independent paths. Considering the downlink channel as reference, transmit and receive spatial correlation refer to the space and ground segment of the system, respectively. A recent ITU-R Recommendation on differential rain attenuation [72] includes an expression for calculating the *rain attenuation spatial correlation coefficient* between two earth stations in communication with the same satellite as a function of their separation distance D (in km), which is plotted in Fig. 7. The expression is valid for temperate mid-latitude areas. From the figure, it is deduced that, to achieve in practice independent fading in the alternative slant paths, earth stations (antennas) should be separated by large distances (> 100 km). On the other hand, transmit spatial correlation refers to the degree of rain attenuation decorrelation achieved by spacing apart two satellites on the geostationary orbit by θ degrees. Expressions for this coefficient can be found in analytical models, one of which [73] has been employed in Fig. 8.

Apart from these independent spatial and temporal correlation models described hitherto, the ideal situation would be to have a space-time description of the satellite channel over the whole satellite coverage. Relevant stochastic models have been presented in [74], [75] and are termed *rain fields*;

⁷Usually, the reception of the same signal from two satellites on ground is referred to as OD in the frame of FS systems and as SatD in the frame of MS systems.

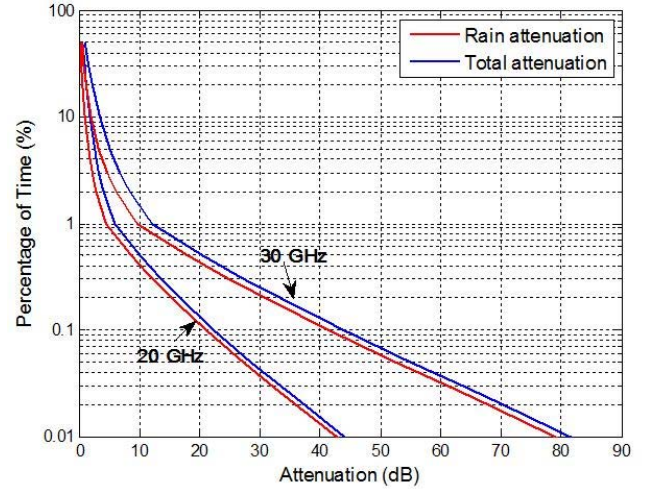


Fig. 5. Rain and total attenuation vs. percentage of time for a Ka band satellite link operating in Singapore.

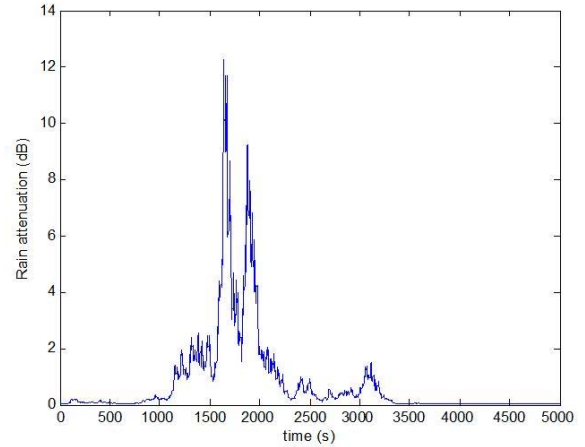


Fig. 6. Snapshot of Ka band attenuation time series synthesis.

however, they fail to provide a high resolution simultaneously in both domains. Taking this into consideration, the authors in [76] introduced the concept of CAs, i.e. the geographical zones within which channel conditions are highly correlated at a given time slot. All FSTs within the same CA experience similar rain fading, whereas FSTs located in different CAs undergo uncorrelated attenuation.

B. Mobile Satellite

Channel characteristics affecting radiowave propagation in the frame of MS systems give rise to an entirely different channel modeling compared to the FS since two fundamental characteristics are completely different: (i) the introduction of *user mobility* (ii) the use of *lower frequency bands* instead of bands above 10 GHz. Concerning the former characteristic, propagation conditions and link geometry are no longer static, insinuating that NLOS communication with the satellite due to heavy shadowing is a strong possibility, especially in urban environments under low elevation angles. Apart from possible degradation of the direct signal from the satellite to the MST,

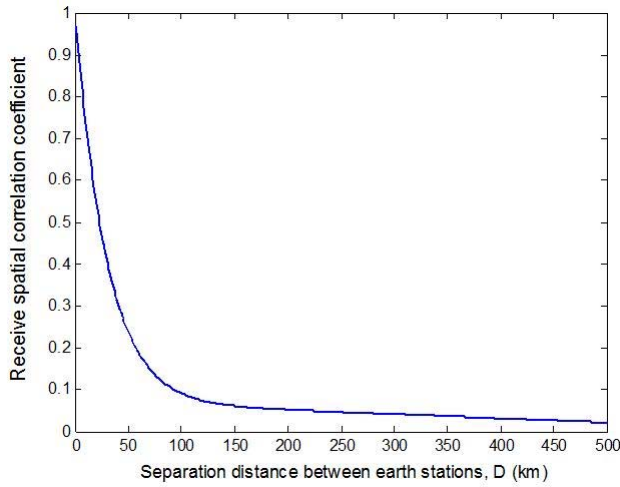


Fig. 7. Receive correlation coefficient of rain attenuation as a function of distance according to the model included in [72].

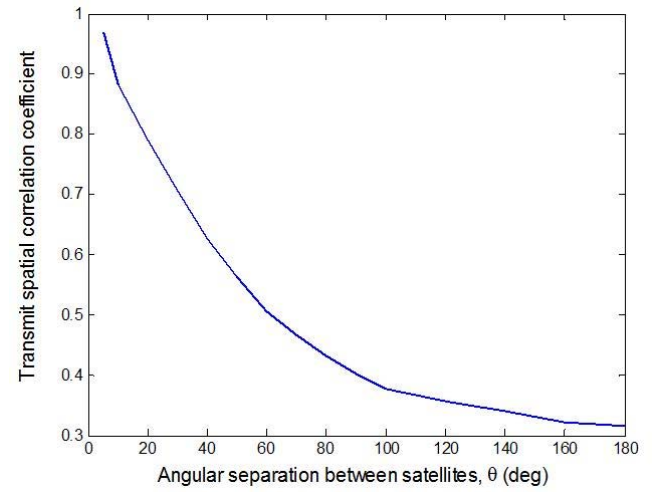


Fig. 8. Transmit correlation coefficient of rain attenuation as a function of the angular separation between two satellites according to the model presented in [73].

the presence of nearby scatterers produces diffuse multipath propagation, which is not present in the LOS, highly directive antenna FS scenario. On the other hand, transmission at the L, S frequency bands instead of Ku and Ka renders tropospheric phenomena irrelevant.

In general, the MS channel comprises two main signal components that should be taken into consideration in the course of MIMO modeling. These include [77]:

- the direct (or LOS) signal;
- the diffuse multipath (or NLOS) component due to the direct signal interacting with the scatterers in the vicinity of the MST;

In turn, given these channel elements and depending on the environment the MST is operating (urban, suburban, rural), the MS channels may be classified according to:

- the degree of time dispersion (narrowband versus wideband);
- the rate of signal variations (very slow, slow and fast variations);
- the combination of statistical distributions (single state models versus multistate models);
- the Doppler power spectrum;
- the spatial correlation.

Time Dispersion: The majority of MS channel models focus on narrowband channels, i.e. they assume a small time dispersion, since, as the experimental evidence indicates [78], this is the most usual situation encountered in practice. Wideband elements can be modeled by a tapped delay line model [79], where each tap is described by a corresponding narrowband model. Hence, narrowband models constitute the basic building blocks for wideband models.

Rate of Signal Variations: As illustrated in Fig. 9, three states of variations of the received signal can be observed: fast, slow and very slow, corresponding to the effects of multipath, shadowing and large scale propagation environment, respectively. In particular, multipath fading is due to scattering in the vicinity of the MST and gives rise to constructive and destructive addition of signals coming

from the multiple paths taking place within a fraction of the wavelength. On the other hand, shadowing is a much slower fading process directly related to the size of a single large or multiple grouped obstacles in the environment near the MST. Nevertheless, when the MST travels over a large area —e.g. when a mobile travels from an open area to an urban area—, shadowing and multipath may change abruptly. Typically, these changes in the large scale propagation environment are modeled by different propagation states, e.g. "good" and "bad" states corresponding to LOS/open/light shadowing areas and NLOS/blocked/heavy shadowing areas, respectively [80], [81]. These propagation states are then described by a first order Markov chain with specific state and transition probabilities. This type of models are called *multistate* MS channel models. Actually, measurement campaigns showed that a refinement of the two-state Markov approach is possible by employing three states, an approach that improves the quality of channel prediction [82], [83], [84], [85]. The third state is the result of splitting one of the two extremes into a third intermediate state to account for moderate shadowing. In an effort to definitely consolidate multistate models, the authors in [86], [87] return to a two-state, where the states do not necessarily correspond to the LOS and NLOS conditions, but are more generic and take on a wider range of possible parameters, which are regenerated for every new state. This reduction in the number of states is quite convenient when modeling MIMO MS in more than one satellite paths.

A possible weakness of the Markov model is that the duration spent at any state must follow the exponential distribution. According to [85] this may result in unrealistic durations, since measurements indicate a minimum state duration of a few meters. Alternatively, [85] proposes a three state semi-Markov model, where the duration or, equivalently, the distance covered in an open area follows a power law, while the state durations under shadowing or blockage follow a lognormal law. The same approach has also been adopted by ITU-R through its Recommendation P.681 [88], which is targeted at MS channel modeling.

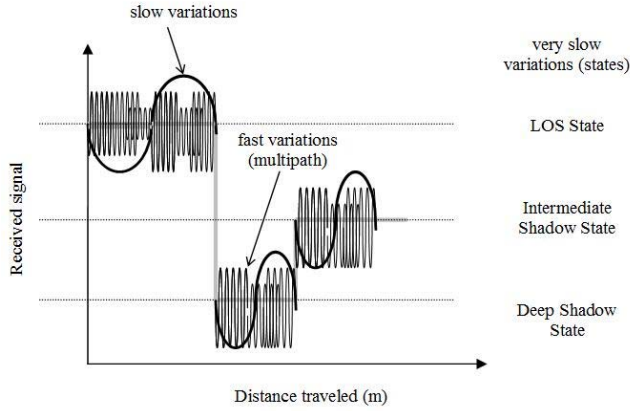


Fig. 9. Different rates of signal variations of the MS channel.

Combinations of Statistical Distributions: The large number of statistical distributions or combinations thereof proposed is due to the fact that a single distribution is not sufficient to characterize the narrowband MS channel. A list of single state and multistate models proposed in the literature is given in Tables I⁸ and II⁹, respectively.

Doppler Spectrum and Simulation of the MS Channel:

The simulation of any MS channel model is carried out by producing time series of the received signal through a so-called *time series synthesizer* [98]. As also stressed in the FS case, these synthesizers are valuable tools to study the channel dynamics and also aid in the design of FMTs. A circuit implementation of the Loo distribution is presented in Fig. 10 [98]. This representation not only considers the reproduction of the envelope of the received signal (magnitude of complex phasor), but also takes into consideration the fast -upper rail of Fig. 10- and slow -lower rail of Fig. 10- fading components.

Spatial Correlation: So far, single satellite links have been considered in this section. In contrast, a number of studies exist on SatD, a technique that improves link availability by employing multiple satellites and its performance is directly related to the degree of correlation between the alternative slant paths. The approach followed in [99] to evaluate the spatial correlation coefficient was to use circular scans within a given environment to obtain numerical landscape pictures, in which a '0' or a '1' would represent link obstruction or visibility, respectively. Furthermore, the two-state Markov model [80] was extended to a four-state Markov model in [100] to model two correlated links. An approach similar to the previous one is presented in [82], where a three-state Markov model is assumed to account for the large dynamic range of the received signal under SatD. In [96] a methodology based on taking fisheye photographs at

⁸The first row of Table I, that is the sum of a Rayleigh and a lognormal process or, equivalently, a Rice distribution conditioned on a lognormal process is widely known as the Loo distribution. Also, the distribution in [93], which includes both the Loo and the Rice-lognormal distributions is widely known as the generalized Rice-lognormal distribution.

⁹The Suzuki distribution refers to a Rayleigh-lognormal distribution of the received signal envelope.

TABLE I

LIST OF MULTISTATE NARROWBAND MS CHANNEL MODELS.

Ref.	Multipath fading	LOS shadowing	Multiplicative shadowing
[89]	Rice	lognormal	-
[90]	Rice	-	lognormal
[91]	Nakagami	-	lognormal
[92]	Beckmann	-	lognormal
[93]	Loo	-	lognormal
[94]	Rice	-	Nakagami

TABLE II

LIST OF SINGLE STATE NARROWBAND MS CHANNEL MODELS.

Reference	Light shadowing	Moderate shadowing	Heavy shadowing
[80]	Rice	-	Suzuki
[81]	Rice	Rice-lognormal	Rayleigh
[95]	Rice	-	Loo
[96]	Rice	Loo	Loo
[82]	Rice	Loo	Rayleigh
[83], [84]	Loo	Loo	Loo
[97]	Rice	Rice	Suzuki
[88]	Nakagami-Rice	Loo	Rayleigh
[86], [87]	Loo	-	Loo

potential user locations and extracting path-state information (clear/shadowed/blocked) from the images as a function of look angles was employed. Finally, an advantage of the SatD methodology presented in [101] and later adopted by [88] over measurement or fisheye campaigns is that it can be applied via a computer simulation based on simulating different realizations (azimuth scan series) of an ON/OFF random process, where the ON and OFF states correspond to the existence of LOS or not.

IV. POTENTIAL APPLICATION OF MIMO OVER SATELLITE

This section contains the main contribution of the article, i.e. the various proposals on how to form a SIMO, MISO or MIMO satellite system having operating characteristics that are relevant to the specifications of the FS and MS study cases.

A. Fixed Satellite

Single-User/Single Satellite MIMO Techniques: The prerequisite so that SU-MIMO configurations fully exploit the diversity and spatial multiplexing advantages predicted by information theory is the existence of a rich scattering environment which renders the fading paths between the multi-antenna transmitter and receiver independent¹⁰. Otherwise, the channel matrix becomes rank deficient and the MIMO system performance degenerates to that of the corresponding SISO system. This is the case of fixed, single satellite systems operating above 10 GHz due to the following reasons:

- 1) the necessity for LOS communication between the satellite and the earth station, which, as already explained, results in a strong spatial correlation and
- 2) the space limitations on board the satellite¹¹.

¹⁰See Section II.A

¹¹Mass and power limitations on board the satellite are also important.

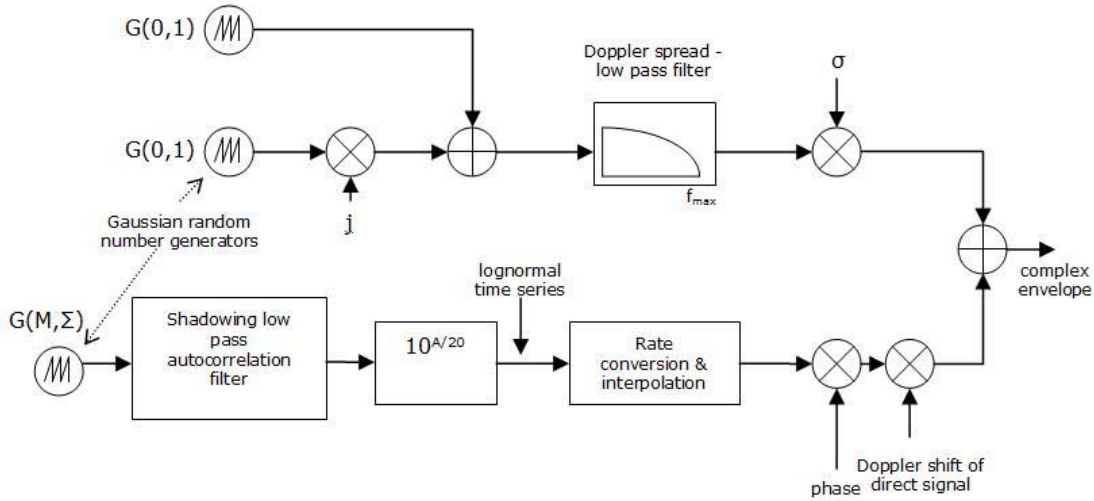


Fig. 10. Circuit implementation of the Loo model with Doppler shaping [98].

Hence, it turns out that a single satellite alone cannot provide the necessary antenna spacings required by MIMO theory. That is why for a SU, single satellite case, efforts have been focused instead on forming a SIMO channel where the antenna multiplicity is exclusively realized on the ground segment by placing two or more earth stations at appropriate distances. SD, implementing receiver selection combining between GW or feeder stations to combat severe rain fading, has been a well known FMT technique since the 70s (see, indicatively, [102], [103], [104]). In this course, the authors in [105] translate the usual SD analysis into a SIMO outage capacity analysis setting the theoretical capacity bounds achieved by a single satellite/dual earth station configuration operating at the Ku band and above.

Single-User/Dual Satellite MIMO Techniques: The space restrictions inherent when using a single satellite turned research to the investigation of dual satellite configurations as a means of profiting by MIMO technology in SU FS communications. Similarly to SD, OD, i.e. the reception by a single earth station equipped with two directional antennas of the same information-bearing signal from two geostationary satellites at a certain angular separation, has been known for long as an FMT against rain fading [106], [107], [108]. Its main drawbacks are the waste of the limited satellite bandwidth for the transmission of the same signal and the need for synchronization of transmission from the two satellites.

Various proposals have been made to investigate the applicability of MIMO-like techniques in configurations comprising two satellites and two or more ground antennas. Motivated by the work in [28], where full MIMO capacity is achieved through a particular geometric arrangement of antennas in terrestrial LOS channels, the authors in [109], [110] propose a two satellite scenario (i.e. the number of transmitting antennas $M_T = 2$), where the receiving antennas are arranged on ground through a geometrical optimization process. For fixed orbital positions of the two satellites and fixed location of the GW station, the number of receive antenna elements

M_R , their in between distances as well as their geometrical arrangement¹² are evaluated. A similar geometrical approach for the optimal placement of antennas in LOS, but for a HAP system [111] has been analyzed in [112]. In [113], assuming two GW stations, two satellites and two FSTs, the emphasis is placed on dealing with the strong correlation of the LOS channel and on suppressing the propagation delay difference between the signals from the two satellites in the single carrier mode (asynchronous reception). According to the authors, these effects are mitigated by a receiver that consists of three modules: (a) a matched filter channel estimator that a priori calculates the delay offset, (b) a spatial MMSE filter to deal with the adjacent satellite interference and (c) a PIC/MRC multi-stage interference canceller. As to the latter module, it is advocated that since the spatial correlation between the propagation paths is quite large, MMSE may prove insufficient to completely cancel interference; hence, the multi-stage PIC is implemented exactly to eliminate any residual interference. Moreover, MRC is used to extract diversity gain from the two receive antennas.

The drawback of all approaches described in the previous paragraphs is that they consider the satellite channel as an AWGN channel ignoring any type of fading. In contrast, the dual satellite/dual earth station MIMO analysis in [114] is carried out assuming both Rayleigh flat fading and rain fading. Comparisons amid the corresponding MIMO, SIMO and SISO configurations with regard to BER performance demonstrate the potential gains achievable by employing two GSO satellites. A statistical investigation of the possible capacity improvement achievable at frequencies above 10 GHz when using a dual satellite, diagonal MIMO system is carried out under correlated rain fading in [115], [116], [117]. Assuming no CSIT, the MIMO capacity under the above assumptions is given by:

$$C = \sum_{i=1}^2 \log_2 \left(1 + \frac{SNR_i}{2} \cdot 10^{-A_{R_i}/10} \right), \quad (13)$$

¹²Assuming a uniform linear array.

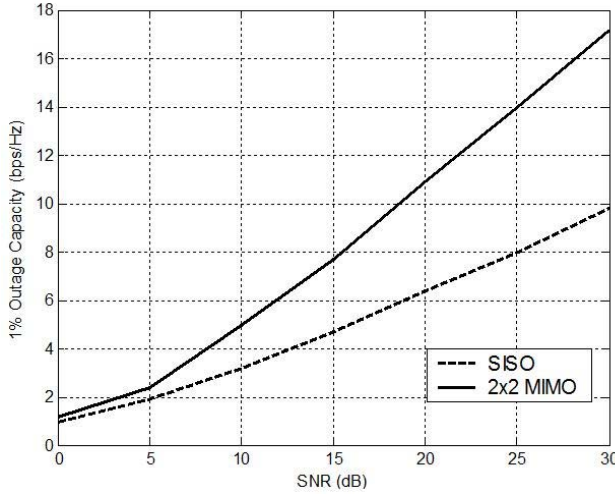


Fig. 11. 1% outage capacity versus SNR for a dual satellite MIMO system compared to the corresponding SISO system [116].

where SNR_i , $i = 1, 2$, denotes the SNR received on ground by each of the two satellites under clear sky conditions and A_{R_i} , $i = 1, 2$, are the random variables representing rain attenuation in dB. Due to A_{R_i} , the capacity C is also a random variable; therefore the proper metric for its assessment is the outage capacity. Fig. 11 reproduces the results of [116] for the 1% outage capacity of a 2×2 MIMO and of the corresponding SISO system revealing the significant advantage offered by the MIMO approach. The system scenario assumed is operation in the Ka band downlink frequency from two GSO satellites with a sufficient orbital separation (40°) under a clear-sky SNR value of 10 dB and specific climatic conditions (Atlanta, GA, US).

Multi-User/Single Satellite MIMO Techniques: The previous two paragraphs manifested that, for a LOS channel suffering from tropospheric fading, applying any kind of SU-MIMO technique is a difficult task, either because space limitations on board a single satellite do not allow for adequate antenna spacing or because employing two satellites gives rise to new challenges, such as waste of spectrum, lack of synchronization in reception and high implementation cost. Anyway, a broadband fixed interactive multibeam satellite system accommodates a large number of FSTs within its multiple beams. Moreover, a direct analogy exists between the forward link of a multibeam satellite channel and the BC of a MU-MIMO system. Indeed, this analogy eliminates many of the disadvantages identified in the SU satellite MIMO channel, since MU-MIMO schemes offer MU multiplexing, are more immune to LOS or antenna correlation and allow for spatial multiplexing gain without necessitating FSTs with multiple antennas. This is especially significant from a commercial point of view, since no modification is necessary to conventional FST receivers.

In practice, instead of the capacity achieving DPC, suboptimal linear and nonlinear MU-MIMO precoding techniques may be employed that sacrifice the sum rate performance for reduced complexity. An initial study applying MU-MIMO precoding in broadband satellite systems was carried out in

the frame of an ESA activity [8]. This work, also summarized in [118], [119], aimed at mitigating interference through linear precoding of the transmit signals. In fact, the scenario considered was the forward link of a transparent multibeam broadband satellite system based on DVB-S2. A GW station was assigned to every cluster of beams and the uplink from the GW to the satellite was assumed ideal. In this context, the GW transmits the vector signal $\mathbf{x} = \mathbf{F} \cdot \mathbf{P} \cdot \mathbf{s}$, where the vector $\mathbf{s}$ is the information to be transmitted with dimension equal to the number of beams in the cluster. $\mathbf{F}$ is the precoding matrix and $\mathbf{P}$ is a diagonal matrix introduced to weight each component of the original signal $\mathbf{s}$ according to a certain criterion. The linear precoding technique chosen was MMSE:

$$\mathbf{F} = (\mathbf{I} + \mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \quad (14)$$

A problem caused by MU-MIMO precoding is that it is common to have more users than beam antennas ($K \gg M$) and, consequently, a user selection rule must be devised. In [119], this is done by employing a TDM strategy, whereby at each time slot only one randomly chosen user per beam is served. Hence, the number of FSTs served becomes equal to the number of antennas M . An issue arises whether fairness among users is achieved, i.e. whether users having different SNR values are chosen with equal probability, which is in contrast with the maximization of the sum rate capacity. This is especially true for GSO satellite networks with wide area coverage (e.g. the whole European continent), since the diverse climatic conditions may give rise to significant fading in certain areas of the coverage and to much smaller in others. In this regard, the *UpConst* algorithm [119] was employed as a compromise between maximizing system throughput and ensuring fairness between FSTs regardless of their location. According to UpConst, the diagonal matrix $\mathbf{P}$ must be:

$$\mathbf{P} = (\mathbf{I} - \text{diag}(\mathbf{SINR})\mathbf{\Phi}^T)^{-1}, \quad (15)$$

where $\mathbf{SINR}$ is the vector of achievable SINR per FST and $\mathbf{\Phi} = \mathbf{H} \cdot \mathbf{F}$. The same work also casts light on the critical issues of *channel estimation* at both the receiver and the transmitter—a necessary prerequisite to profit from MU-MIMO precoding—, as well as on the effect of *transponder nonlinearity*, a problem that is intensified when multiple signals are simultaneously amplified. To deal with the first issue, every FST can estimate a single row of $\mathbf{H}$ employing the unique word at the SOF. Next, every FST feeds this information back to the GW, which can then construct the whole $\mathbf{H}$ matrix to be used for precoding. To deal with the transponder nonlinearity, it is concluded that the OBO must be increased compared to single-carrier amplification for the purposes of precoding. As for the performance of the proposed system, comparative results concerning the implementation of a linear precoding scheme through 11 clusters of 8 beams and through a single cluster of 88 beams are presented in Table III¹³ in terms of availability and total throughput. The benchmark scenario without precoding is also shown as a reference.

¹³Each cluster is assigned with a GW that manages traffic and resource allocation.

TABLE III
PERFORMANCE OF THE MU-MIMO LINEAR PRECODING SCHEME
PROPOSED IN [119].

	Linear Precoding Scheme		No Precoding
	11 clusters of 8 beams	Single cluster of 88 beams	Benchmark system
Availability (%)	84.63	98.53	99.887
Throughput (Gbps)	22.4	28.8	18.55

An improvement over the previous MMSE precoding scheme was recently presented in [120], again relying on the same system characteristics as the FS study case under examination. The work was motivated by the opportunistic beamforming concept¹⁴, which was introduced in [57] in the frame of SU-MIMO and was later extended to MU-MIMO [53] as an interference mitigation technique. The authors in [120] go a step further than MOB by proposing an IMOB technique for the generation of the $M \times M$ precoding matrix $\mathbf{F} = [\mathbf{f}_1 \cdots \mathbf{f}_M]$, where $\mathbf{f}_m$, $m = 1, \dots, M$, is the $M \times 1$ precoding vector selected by each user. They argue that the random generation of $\mathbf{f}_m$ does not extract all the benefits of MIMO processing in FS. Instead, the precoding vectors can be generated taking into account the spatial power density at each fixed ground location, i.e. the a priori knowledge of each FST position with respect to the beam center. Concerning the feedback load necessary for IMOB, only the progressively selected users are asked to provide their full CSI. The precoding vectors are sequentially generated according to a step-by-step procedure detailed in the paper that provides triangular interference cancelation for the selected users. Since the channel estimation procedure is also done iteratively, the feedback to the GW necessary for the IMOB requires M SINR values and M complex values, i.e. slightly more overhead than MMSE precoding and the typical MOB.

Coming now to the performance of the IMOB scheme, assuming a cluster of $M = 7$ beams and typical system characteristics, Table IV [120] presents a performance comparison between a reference SISO system, MMSE, MOB and IMOB in terms of rate, and availability for a Ka band satellite with EIRP equal to 72 dBW. For every technique, results are given for beam 1 –the best performing beam– and beam 4 –the cluster’s center beam suffering more from interference–, except for the IMOB, where the worst performing beam is 7 (last in the iterative algorithm). It is observed from this table that, when the MOB precoder is applied, the orthogonality is lost during propagation, albeit the corresponding precoding matrix is generated so that its columns are orthogonal. The IMOB outperforms MMSE in all system aspects, however, sacrifices to an impractical extent availability in some beams.

Precoding techniques examined hitherto were linear, characterized by computational simplicity. Nonlinear precoding involves additional transmit signal processing to improve the error rate performance; however, some schemes like THP

TABLE IV
PERFORMANCE COMPARISON OF VARIOUS SATELLITE MULTIBEAM
PRECODING TECHNIQUES [120].

		Rate (bps/Hz)	Availability (%)	Rate Variance
Reference	Beam 1	2.55	96.3	1.35
	Beam 4	1.45	92.7	0.16
	Aggregate	16.80	95	1.19
MMSE	Beam 1	3.16	84.9	4.24
	Beam 4	1.89	74.8	1.63
	Aggregate	20.90	83.7	3.89
MOB	Beam 1	0.86	43.0	0.11
	Beam 4	0.86	42.5	0.11
	Aggregate	6.04	42.7	0.11
IMOB	Beam 1	8.09	100	3.74
	Beam 7	2.19	87.6	0.74
	Aggregate	24.40	95.5	1.12

[54] have been proven practical to some extent. This was acknowledged in [121], where THP was implemented for the forward link of a broadband multibeam satellite system and was compared to its linear counterpart. Three THP versions are considered in this paper, mainly differing in the respective definition of the matrix $\mathbf{G}$ (see Fig. 4):

- The ZF-THP scheme proposed in [54], where $\mathbf{G} \neq \mathbf{I}$ and, consequently, each user faces a different SNR level.
- The ZF/MMSE-THP scheme proposed in [122], where $\mathbf{G} = \mathbf{I}$ and, consequently, SNR is equally balanced among all users.
- The ZF-THP scheme proposed in [123]. It can be considered as a generalization of the previous scheme, allowing different QoS levels (i.e. SNR) to each user.

Under a scenario where the GW controls 10 co-channel beams using QPSK, BER results show that THP outperforms linear precoding in the high SNR region. However, for SNR values below a certain threshold, MMSE linear precoding performs better than ZF-THP. In a scenario where the GW controls 10 beams, but a three colour frequency plan is applied (resulting in reduced interference) using 16QAM, precoding offers almost no advantage, except for very high SNR values.

Finally, an interesting investigation of dual polarization for MU-MIMO processing in multibeam FS systems is carried out in [124]. Inspired by the space limitations of a single GSO satellite, the authors turn to dual polarization and compare throughput and availability for a number of multiplexing/diversity techniques:

- *Best SNR polarization selection.*
- *Random polarization selection.*
- *Stream multiplexing*, where the information toward a single user is transmitted through two parallel streams (each over a given polarization) and encoding and decoding are carried out independently.
- *Multiplexing with SIC decoder*, where SIC is applied at the receiver to mitigate the interference caused by the two polarizations (joint decoding).
- *SVD multiplexing*, where joint processing both at the receiver and the transmitter is carried out.
- *STBC*, where the Alamouti transmit diversity scheme [32] is applied.

¹⁴See Section II.

TABLE V
PERFORMANCE COMPARISON OF SATELLITE MULTIBEAM DUAL
POLARIZATION TECHNIQUES [124].

		Rate (bps/Hz)	Availability (%)
Single polarization	Beam 1	2.59	96.9
	Beam 4	1.46	92.2
Best SNR selection	Beam 1	4.29	100
	Beam 4	2.66	100
Random selection	Beam 1	4.36	100
	Beam 4	2.67	100
Independent multiplexing	Beam 1	2.99	97.8
	Beam 4	0.92	92.2
Joint decoding (SIC)	Beam 1	3.00	97.8
	Beam 4	0.92	92.2
Joint encoding/decoding (SVD)	Beam 1	0.05	0
	Beam 4	0.07	0
STBC (Alamouti)	Beam 1	2.59	96.9
	Beam 4	1.46	92.2

The results from simulations of these strategies for typical satellite and FST parameters are displayed in Table V [124] (for an antenna XPD equal to 30 dB). The final conclusion drawn by the authors is that dual polarization in a fixed multibeam satellite scenario can only serve as an additional colour (i.e. degree of isolation) in the frequency planning process.

Considering the incorporation of any MIMO precoding scheme in the DVB-S2 standard, there are considerations regarding the impact on the *encapsulation efficiency* of the protocol. To ensure a high encapsulation efficiency in DVB-S2, each ACM mode shall include packets addressed to various users. However, as [120] revealed, a significant inconsistency with respect to the ACM function of the standard exists. In particular, it is possible that data packets destined to different FSTs are encapsulated in the same BBFRAME with a single ACM mode [125]. Although this is an efficient strategy for non-MIMO systems, it limits the implementation of MU-MIMO precoding as all the SINR values within the same beam must be similar¹⁵ in order to be included in the same ACM format. However, whenever the spatial precoding matrix $\mathbf{F}$ is modified, the SINR of the FSTs also changes due to change in the interference between the users. This means that users in the same BBFRAME may not have a comparable SINR and that the ACM module should modify the ACM mode within the BBFRAME, an action that is prohibited by DVB-S2. Unfortunately, this observation seems to limit the application of precoding only to schemes that serve FSTs of comparable SINR. As a result, besides throughput and availability which are usually employed as performance metrics, [120] introduces a third one, namely the *rate variance* within the BBFRAME. The smaller this variance is, the higher the probability that all FSTs included in the same BBFRAME use the same ACM mode¹⁶.

¹⁵Lie within the bounds of the same SINR region for a specific ACM mode [126].

¹⁶Results of the rate variance are shown in the third column of Table IV.

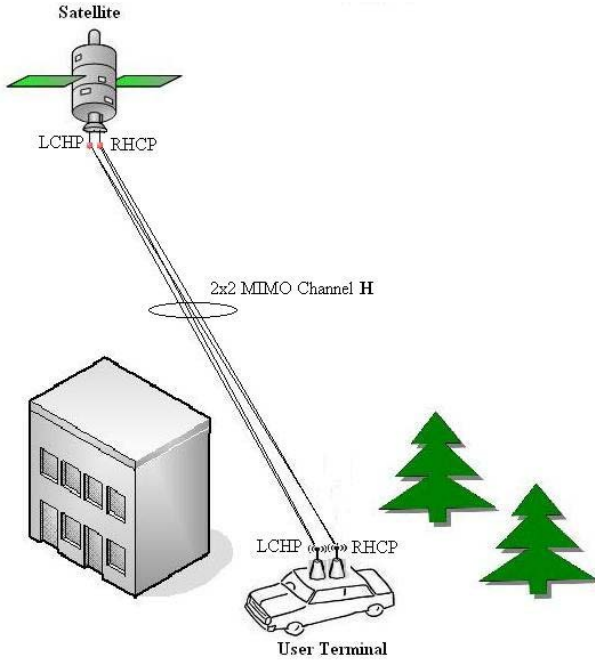
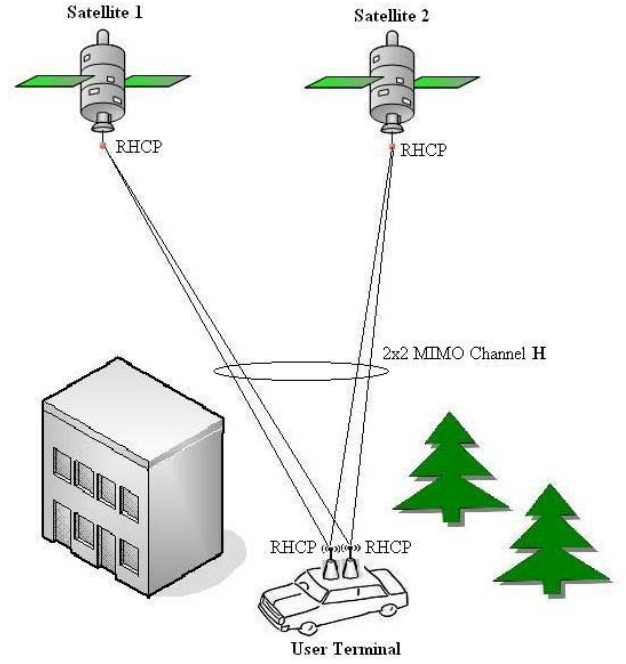
B. Mobile Satellite

MIMO Land Mobile Satellite Measurements and Channels: The overwhelming majority of analytical and experimental works carried out on MIMO MS channels at the L and S frequency bands focus on exploiting polarization diversity at both the transmitter and receiver to form a 2×2 MIMO matrix. This approach, depicted in Fig. 12, has been found extremely beneficial since a single satellite cannot provide the necessary antenna spacings required by MIMO theory to provide a high degree of channel decorrelation.

Spatial, temporal and polarization SIMO and MISO measurements in S and C bands for MS have been recently carried out by ESA employing existing satellites [127] and by CNES employing a helicopter [128]. A relevant SIMO channel modeling approach is presented in [129]. With regard to MIMO measurement campaigns, the relevant attempts are extremely scarce and have been conducted mainly in the frame of [130] in Guildford, UK, at 2.45 GHz. To emulate the satellite scenario, an artificial terrestrial platform acting as the satellite was installed on a hilltop transmitting to a mobile vehicle (van) acting as a MST. This configuration led to low elevation angles ranging between 5° and 18° . Three propagation environments were measured: (a) a tree-lined road, (b) a suburban area and (c) an urban area. The vehicle moved at a velocity of 8.9 m/s in the tree-lined road and 5.6 m/s in the suburban and urban environments. Furthermore, two satellite geometries were emulated to form a 2×2 MIMO matrix:

- 1) a dual satellite scenario (spatial diversity, see Fig. 13) by installing two antennas on the hilltop at a distance of 10 wavelengths and employing two antennas on the MST at a distance of 4 wavelengths [131], [132]. All antennas employ the same polarization.
- 2) a single satellite/dual polarization scenario (polarization diversity, see Fig. 12) by separating the two antennas on the hilltop only by 1 wavelength and the two antennas on the MST again by 4 wavelengths [133], [134]. LHCP/RHCP pairs of antennas were used.

For the dual satellite scenario in [132], the authors first derive the normalization procedure of the 2×2 MIMO channel matrix $\mathbf{H} = A\hat{\mathbf{H}}$, where $\hat{\mathbf{H}}$ is the measured matrix and $A = 2/\|\hat{\mathbf{H}}\|_F$. Then, the measured channel matrices are substituted into (2) setting $\mathbf{Q} = \mathbf{I}_2$ –MIMO capacity formula when no CSIT is available– and the outage capacity for the experimental 2×2 MIMO system is calculated. The results show a significant improvement of the outage capacity, e.g. for the 10% outage capacity, 0.06, 0.06 and 0.12 bps/Hz for SISO and 0.12, 0.12 and 0.24 bps/Hz for MIMO in the main road, suburban and urban environment were found, respectively. Following the same line of work but employing a dual polarization MIMO system, results in [133] suggest that 10% of the channels are greater than 0.02, 0.09 and 0.03 bps/use for SISO and 0.14, 0.37 and 0.26 bps/use for MIMO in the main road, suburban and urban environments, respectively. The small-scale properties of the wideband MIMO channel are revealed after subtracting the average power of the large-scale fading from the measured data. The statistics of the small-scale fading conditioned on the large-scale shadowing was approximated by the Rice distribution.


 Fig. 12. System configuration of a 2×2 dual polarization MIMO MS system.

 Fig. 13. System configuration of a 2×2 spatial MIMO MS system.

As expected, the scarcity of MIMO MS channel measurements is responsible for the corresponding scarcity of proposed MIMO MS channel models in the literature. Some preliminary efforts to extend the large amount of work on MIMO channel modeling of dual polarized terrestrial systems to the satellite scenario have been carried out in [130], [135]. An analytical statistical channel model for narrowband MIMO MS channels is also presented in [136]. In order to ease the tractability of the mathematical analysis, this MIMO channel model assumes that the envelope of the large-scale fading components is Nakagami distributed (thus its power is Gamma distributed), the signal envelope of the small-scale fading components is Rayleigh distributed whereas no correlation issues among these components are addressed.

Interestingly, the authors in [137] make use of the consolidated channel model presented in [86], [87] for SISO DVB-SH systems to derive a consolidated channel model for MIMO DVB-SH like systems. Its main application concerns narrowband MIMO MS channels employing dual polarization; hence, the dimension of the channel matrix $\mathbf{H}$ is 2×2 . Similarly to [86], [87], the signal amplitude is assumed to follow the Loo distribution. According to (5), the channel matrix $\mathbf{H}$ can be expressed as the sum of a channel matrix $\tilde{\mathbf{H}}$ accounting for shadowing and of a channel matrix $\hat{\mathbf{H}}$ accounting for small scale fading, where the magnitudes $|\tilde{h}_{ij}|$ are lognormally distributed with parameters (α, ψ) and the magnitudes $|\hat{h}_{ij}|$ are Rayleigh distributed with MP expressing the multipath power. The triplets (α, ψ, MP) are determined for every type of environment according to [84]. The spatial correlation of the large-scale fading component is given by:

$$\text{vec}(\tilde{\mathbf{H}}) = 10^{(\psi/20)} \tilde{\mathbf{C}}^{1/2} \cdot \text{vec}(\tilde{\mathbf{H}}_w) + (\alpha/20), \quad (16)$$

where $\tilde{\mathbf{H}}_w$ is the 2×2 channel matrix with spatially uncorrelated ZMSCG elements of zero mean and unit variance and $\tilde{\mathbf{C}}$

is the 4×4 positive semi-definite Hermitian covariance matrix of the large-scale fading components according to [130]. The XPD of the large-scale fading components $\tilde{h}_{ij}$ is related only to the ability of the antennas to separate the two orthogonal polarizations, XPD_{ant} , that is:

$$10 \log_{10} \left(\frac{E[|\tilde{h}_{ii}|^2]}{E[|\tilde{h}_{ij}|^2]} \right) = 10 \log_{10} \left(\frac{1 - \beta_{ant}}{\beta_{ant}} \right) = XPD_{ant} \quad (17)$$

On the other hand, the spatial correlation of the small-scale fading component is given by (4). The model assumes symmetry of the co- to cross-polarization ratio independently of the reference polarization, hence:

$$E[|\tilde{h}_{ii}|^2] = MP \cdot (1 - \gamma) \quad i = 1, 2 \quad (18)$$

$$E[|\tilde{h}_{ij}|^2] = MP \cdot \gamma \quad i = 1, 2 \quad (19)$$

where according to the expressions:

$$\gamma = \beta_{ant}(1 - \gamma_{env}) + (1 - \beta_{ant})\gamma_{env} \quad (20)$$

$$XPD_{env} = 10 \log_{10} \left(\frac{1 - \gamma_{env}}{\gamma_{env}} \right) \quad (21)$$

γ depends on both XPD_{ant} and XPD_{env} . Taking the above definitions into consideration, the receive covariance matrix $\mathbf{R}_R = E[\tilde{\mathbf{H}}\tilde{\mathbf{H}}^H]$ and transmit covariance matrix $\mathbf{R}_T = E[\tilde{\mathbf{H}}^H\tilde{\mathbf{H}}]$ of the small-scale fading component are expressed by:

$$\mathbf{R}_i = MP \begin{bmatrix} 1 & 2\sqrt{(1-\gamma)\gamma}\rho_i \\ 2\sqrt{(1-\gamma)\gamma}\rho_i & 1 \end{bmatrix} \quad i = T, R \quad (22)$$

where ρ_T, ρ_R are the small-scale fading spatial correlation coefficients at the transmitter and the receiver, respectively. Fig. 14 presents an output of this MIMO MS channel model for S band dual polarization MIMO in the open, suburban and

urban areas. The time series have been produced assuming an MST velocity of 50 km/h.

Space-Polarization-Time Coding Techniques: The literature review of MIMO techniques that are relevant to MS is classified into single satellite/dual polarization techniques -referred to as PTC, see Fig. 12- and dual satellite techniques -referred to as STC, see Fig. 13-. As already mentioned, the various proposals for implementing PTC instead of STC MIMO in SatCom are motivated by the lack of synchronization in SatD configurations. Though in terrestrial systems the relatively short distance between the transmitter and the receiver makes the decoding of the received signal a rather simple task, in the SatD case, the satellites are very far from each other. As a consequence, the arrival times of signals originating from two satellites (forming part of a single code word) may be shifted by tens or hundreds of symbol times relative to each other [138].

In [139] a basic investigation of PTC was presented using a simple 2×2 MIMO channel model employing two orthogonal circular polarizations (RHCP/LHCP) and assuming that in a multipath environment the received signal is of random polarization. As a result, the application of the Alamouti scheme [32] produces diversity gain, which depends on the degree of correlation. The extension of this work to 3D polarization [138], [140], where the random polarization orientation under multipath propagation is considered a 3D phenomenon. Hence, the number of orthogonal polarization states is three, increasing correspondingly the number of orthogonal channels to three, a fact first highlighted in [31]. Since the MIMO channel matrix is at most rank 2, even if the antenna aperture is illuminated by a 3D polarized wave, only two of its components would travel toward the ground terminal, resulting in a two-fold capacity increase achieved by PTC. On the contrary, diversity can take the full advantage of multipolarized transmission: For a 3D polarized satellite antenna and a conventional dual polarized MST antenna, a diversity gain of 6 is possible.

Although the concept of 3D polarization seems attractive, it requires sophisticated antennas, at least at one end of the satellite link. On a more practical level, [141] presents a comprehensive simulation of a MIMO scenario based on dual polarization (RHCP/LHCP) in the frame of the DVB-SH standard. Two PTC schemes are applied: Alamouti coding [32] and STTC [34]. The results of the comparison of these advanced techniques with a SISO satellite system are summarized as follows [141]:

- The 2×2 MIMO version offers improved BER performance compared to simply multiplexing two streams along each polarization. This is true for either STTC or Alamouti.
- STTC exhibits an equal or better BER than the SISO case and also doubles the spectrum efficiency.
- STTC offers a 1.5, 1, and 0.5 dB gain at speeds of 3, 50, and 120 km/h, respectively, when compared with independent coding techniques.
- The Alamouti scheme offers approximately a 1 dB gain with respect to SISO for all MST velocities.

In [135], a layered PTC technique for MS communications

is combined with IDD at the receiver. This complies with the DVB-SH standard, which also uses a Turbo coding module. Actually, [135] is an extension of the work on Turbo-BLAST carried out for terrestrial MIMO systems [39]. Its main contribution is that it achieves significant coding gains and diversity advantages by using layered PTC based on very powerful, though one-dimensional error correcting codes (i.e. Turbo codes). The MIMO channel model applied in [135] resembles the one in [137] described earlier, differing in the adoption of the Rice instead of the Loo distribution. The PTC strategy of [135] is as follows: At the transmit side the encoded/modulated data are mapped onto two streams and are simultaneously transmitted using two orthogonal polarizations. A disadvantage of the satellite-based version of PTC is that separating the two simultaneously transmitted, independent signals is more difficult in the presence of strong LOS (high K factors). To tackle this problem and, in turn, reduce the effective cross-polarization interference, the authors introduce a *constellation rotation* technique. To elaborate on this technique, let the 2×1 received signal vector $\mathbf{y}$ be expressed in the usual matrix form (1). When the channel is constant for at least L symbols, the received signal in matrix notation can be written as:

$$\mathbf{Y} = \mathbf{H}\mathbf{S}' + \mathbf{N}, \quad (23)$$

where the $2 \times L$ matrix $\mathbf{S}'$ is limited by $\text{Tr}(\mathbf{E}[\mathbf{S}'^H \mathbf{S}']) \leq E_S L$, $\mathbf{Y}$ and $\mathbf{N}$ are also of dimension $2 \times L$ and PTC multiplexing of the $2L$ symbols is carried out according to the matrix:

$$\mathbf{S}' = \begin{bmatrix} s_{11} \exp(j\theta) & \cdots & s_{L1} \exp(j\theta) \\ s_{12} & \cdots & s_{L2} \end{bmatrix} \begin{matrix} \rightarrow \text{polarization 1} \\ \rightarrow \text{polarization 2} \end{matrix} \quad (24)$$

In (24), the first row is transmitted using polarization 1 and the second row is transmitted using its orthogonal polarization. Note that the signal constellation transmitted using polarization 1 is rotated by an angle θ . Equation (23) can be rewritten as:

$$\mathbf{Y} = \mathbf{H}'\mathbf{S} + \mathbf{N}, \quad \mathbf{S} = \begin{bmatrix} s_{11} & \cdots & s_{L1} \\ s_{12} & \cdots & s_{L2} \end{bmatrix}, \quad (25)$$

where the constellation rotation has been applied to the channel matrix:

$$\mathbf{H}' = \begin{bmatrix} h_{11} \exp(j\theta) & h_{21} \\ h_{12} \exp(j\theta) & h_{22} \end{bmatrix} \quad (26)$$

When the elements h_{ij} of the channel matrix $\mathbf{H}$ are Rayleigh distributed—as in the terrestrial case—the angle rotation introduced in the first column of the channel matrix will not reduce any interference. However, when there is a LOS component—as in the satellite case—the rotation of the first column will provide some angular diversity in the attempt to separate the interfering signals. For Rician channels with large K factors, the optimal constellation rotation θ for a QPSK modulated signal is approximately 45° [135].

In all simulations, the performance comparison of the following schemes was made:

- polarization multiplexing employing IDD receivers;
- transmit polarization diversity employing the Alamouti scheme coupled with receive diversity at the receiver;

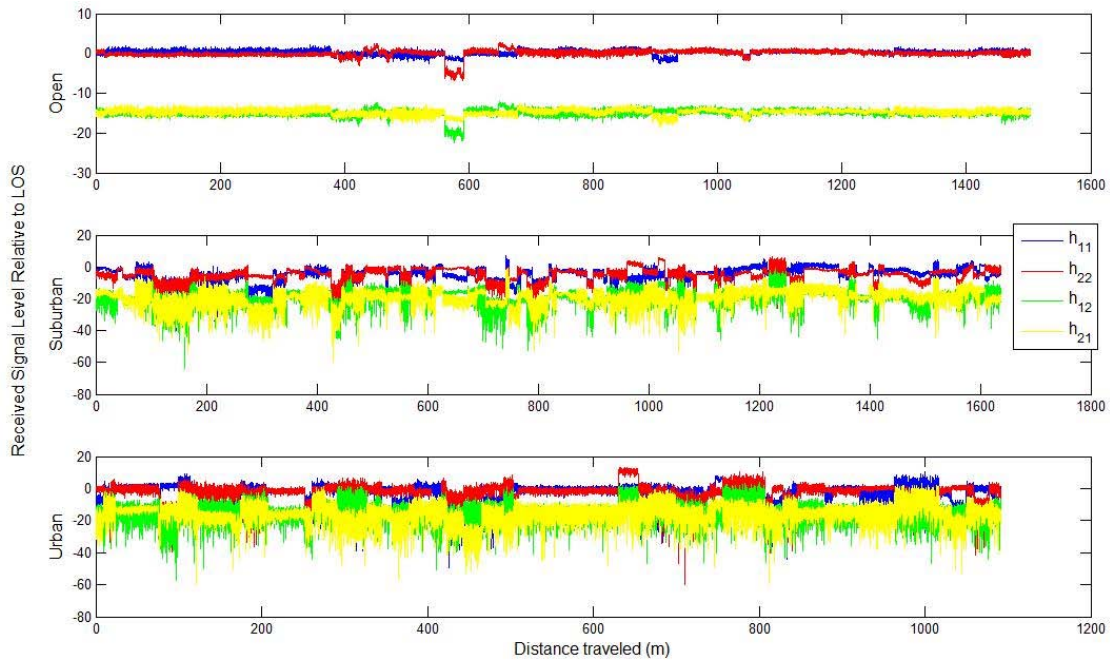


Fig. 14. Time series of S band dual polarization MIMO MS channel model for open, suburban and urban area [137]. Assumed MST velocity is 50 km/h and elevation angle 40° .

- receiver polarization diversity only (a single polarized transmit antenna coupled with a dual polarized receive antenna).
- no polarization diversity (one polarization at the transmit and at the receive sides);

Except for the Alamouti scheme, all techniques use a code rate $1/2$, constraint length five, flushed convolutional code and QPSK modulation. Note that, the spectrum efficiency of the schemes using dual polarized transmit and receive antennas is 2 bps/Hz, whereas the information rate of the schemes without polarization diversity or receive only polarization diversity is 1 bps/Hz. The transmit power was considered fixed in all four configurations. Since [135] presents a rather large number of simulation examples and figures, it is preferable to sum up the observations concerning the BER performance of these experiments as it was done by the authors:

- Due to the unused polarization, the scheme without polarization diversity is inferior to receive only polarization diversity, Alamouti, and polarization multiplexing for only a few iterations (3-5) of the IDD.
- As the number of iterations in the IDD increases, the performance of the iterative decoding improves. Most of the iterative gain is achieved in three iterations.
- Looking solely at the more advanced techniques, the BER performance of the polarization multiplexing/IDD receiver exceeds that of the Alamouti polarization diversity at moderate to high SNR values. The reason is the coding gain achieved by the convolutional codes, whereas the Alamouti scheme offers only diversity and no coding gain.
- However, the BER performance of the receive polarization diversity is better than that of polarization multiplexing at low to moderate SNR values because of its lower information rate. However at the low SNR

region, the BER of the same scheme is worse than that of the Alamouti scheme. That means that, for low SNR, polarization diversity may be a better choice than polarization multiplexing.

One of the few publications that deal with a complete space-polarization-time coding based on a SatD geometry is [141], where two satellites each with dual polarization and one MST with dual polarization are in communication. Such a configuration is shown in Fig. 15. Under this 2×4 MIMO scheme, the 2×1 received symbol becomes:

$$\mathbf{y} = \begin{bmatrix} \mathbf{H}_{sat_1} & \mathbf{H}_{sat_2} \end{bmatrix} \mathbf{s} + \mathbf{n}, \quad (27)$$

where $\mathbf{H}_k$, $k = sat_1, sat_2$, is the 2×2 channel matrix from satellite k to the MST, because of the dual polarization employed. The 4×1 transmit signal vector is written in compact form as:

$$\mathbf{s} = \begin{bmatrix} s_{\parallel}^{sat_1}(n) & s_{\perp}^{sat_1}(n) & s_{\parallel}^{sat_2}(n-d) & s_{\perp}^{sat_2}(n-d) \end{bmatrix}^T, \quad (28)$$

where n corresponds to discrete symbol period index, d is the relative delay introduced due to the length difference two satellite paths and the subscripts $\parallel, \perp$ refer to the two orthogonal polarizations. Moreover, this work deals directly with the lack of synchronization inherent in this scheme by proposing different approaches. One approach consists in independent encoding of the signals from the two satellites using OSTBC in two separate frequency bands (one band per satellite) and applying selection combining at the receiver, i.e., at each sampling period, the band experiencing less fading is chosen. The price for this is that the spectrum efficiency is reduced to half since the same information is transmitted in two frequency bands. Another possibility investigated is the design of time-interleaved OSTBC [142]. A third alternative proposed by the authors is a coarse alignment of the satellite

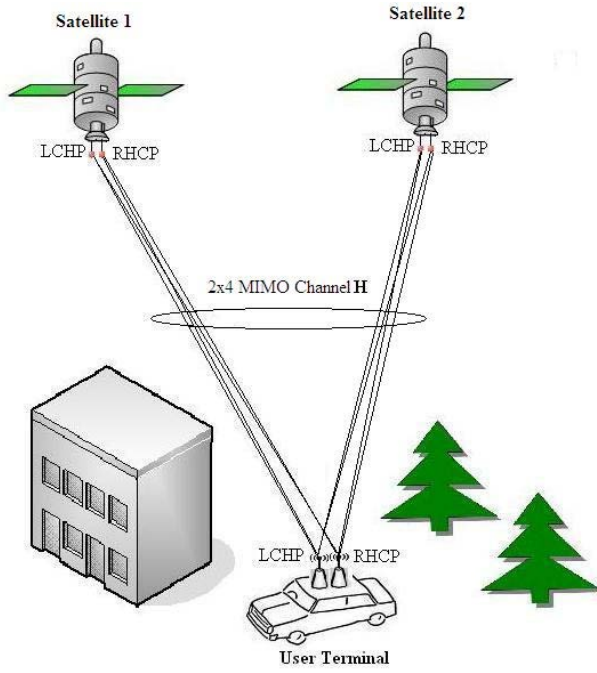


Fig. 15. System configuration of a 2×4 dual polarization/spatial MIMO MS system.

delay. However, the two latter alternatives necessitate a code that is designed for specific delay, which is not applicable in satellite broadcasting applications, where each MST experiences different relative delays. A practical solution the authors in [141] arrive at is a *block-wise Alamouti scheme* [143]. For block sizes of transmitted symbols equal to N and using a GI between the blocks, the block transmission of symbols is schematically explained in Fig. 16. If d_1, d_2 represent the propagation delays from each satellite, the per block received symbol vectors of dimension $2 \times N$ are given by:

$$\mathbf{Y}^{block_1} = \begin{bmatrix} \mathbf{H}_{sat_1} & \mathbf{H}_{sat_2} \end{bmatrix} \begin{bmatrix} \mathbf{s}_{sat_1}(n - d_1) \\ \mathbf{s}_{sat_2}(n - d_2) \end{bmatrix} + \mathbf{N}^{block_1} \quad (29)$$

$$\mathbf{Y}^{block_2} = \begin{bmatrix} \mathbf{H}_{sat_1} & \mathbf{H}_{sat_2} \end{bmatrix} \begin{bmatrix} -(\mathbf{s}_{sat_2}^{rev}(N + GI + d_1 - n))^* \\ (\mathbf{s}_{sat_1}^{rev}(N + GI + d_2 - n))^* \end{bmatrix} + \mathbf{N}^{block_2}, \quad (30)$$

where $\mathbf{s}_{sat_1}, \mathbf{s}_{sat_2}, \mathbf{s}_{sat_1}^{rev}$ and $\mathbf{s}_{sat_2}^{rev}$ are defined in Fig. 16 and $0 \leq n \leq N$. The total $4 \times N$ received vector from the two blocks can be written in a more compact form as:

$$\mathbf{Y} = \begin{bmatrix} \mathbf{H}_{sat_1} & \mathbf{H}_{sat_2} \\ \mathbf{H}_{sat_2}^* & -\mathbf{H}_{sat_1}^* \end{bmatrix} \begin{bmatrix} \mathbf{s}_{sat_1}(n - d_1) \\ \mathbf{s}_{sat_2}(n - d_2) \end{bmatrix} + \mathbf{N} \quad (31)$$

Note that due to the receiver antenna XPD, the cross-polar terms are low. As a result, applying a matched filter receiver results in an almost perfect separation between the signals sent over the orthogonal polarizations. However, although small, ISI cannot be neglected and, therefore, to minimize this interference the two following measures should be taken [141]: (a) To avoid inter-block interference, the GI has to be at least equal to the relative delay $d = |d_1 - d_2|$. In fact, the system efficiency will improve the larger N is compared to d . (b) The length of the frame $2(N + GI)$ should be

less than the channel coherence time. Hence, a tradeoff exists between these two conditions. A second level of tradeoff is due to condition (a), which implies satellites that are closely spaced in orbit –to keep the GI as low as possible– and the fact that closely spaced satellites do not deliver the necessary decorrelation of signals. The comparative simulation [141] takes into consideration the following techniques (the occupied resources by each technique are also quoted):

- 1) Alamouti transmission in two separate frequency bands and selection combining at the receiver (SatD). Occupied bandwidth = $2B$, transmit power = $2P$.
- 2) Block-wise Alamouti with matched filter receiver (SatD). Occupied bandwidth = B , transmit power = $2P$.
- 3) Block-wise Alamouti with ZF receiver (SatD). Occupied bandwidth = B , transmit power = $2P$.
- 4) Alamouti transmission from a single satellite (benchmark scenario). Occupied bandwidth = B , transmit power = P .

Technique 1 occupies double the spectrum compared to the rest, whereas in the block-wise Alamouti scheme (techniques 2 and 3) both satellites share time and frequency. However, to obtain the same BER as technique 1, a SNR penalty of 4 dB is imposed on techniques 2 and 3. The matched filter and ZF receivers exhibit a similar performance because the ISI terms due to cross-polarization are rather low. However, the ZF technique is superior to the matched filter in terms of BER.

To sustain a high degree of coverage even for indoor handheld users in urban areas, the satellite interface in DVB-SH based MS is usually complemented by a network of terrestrial repeaters (gap fillers) [144] forming a hybrid (satellite+terrestrial) wireless interface [145]; the gap filler network is usually set up as a SFN. From a MIMO perspective, the interest in this configuration lies in the possible application of *distributed* or *virtual* MIMO techniques [146]. In this scenario, the satellite and the terrestrial repeater can "share" their antennas and send different parts of a space-time codeword, resembling a MIMO transmitter, though the transmit antennas are geographically dispersed. Particular implementations of a distributed Alamouti scheme are envisaged in [141], [147].

V. CONCLUSIONS & FUTURE RESEARCH PERSPECTIVES

A. Fixed Satellite

The application MU-MIMO precoding techniques on multi-beam satellites seems currently the way for FS systems to profit from MIMO in the short term since it resolves a number of shortcomings appearing in the SU-MIMO case in a cost efficient, practical way (single satellite, single antenna FSTs) based on familiar technology (multibeam satellites). The price that must be paid for these advantages is that MU-MIMO necessitates CSIT, which is translated into some kind of feedback. Moreover, linear and non-linear precoding techniques show a moderate throughput advantage, while penalizing availability in some beams. A serious consideration is also the compatibility with the existing DVB-S2 standard. Possible extensions to previous work may be sought along the lines of the following axes:

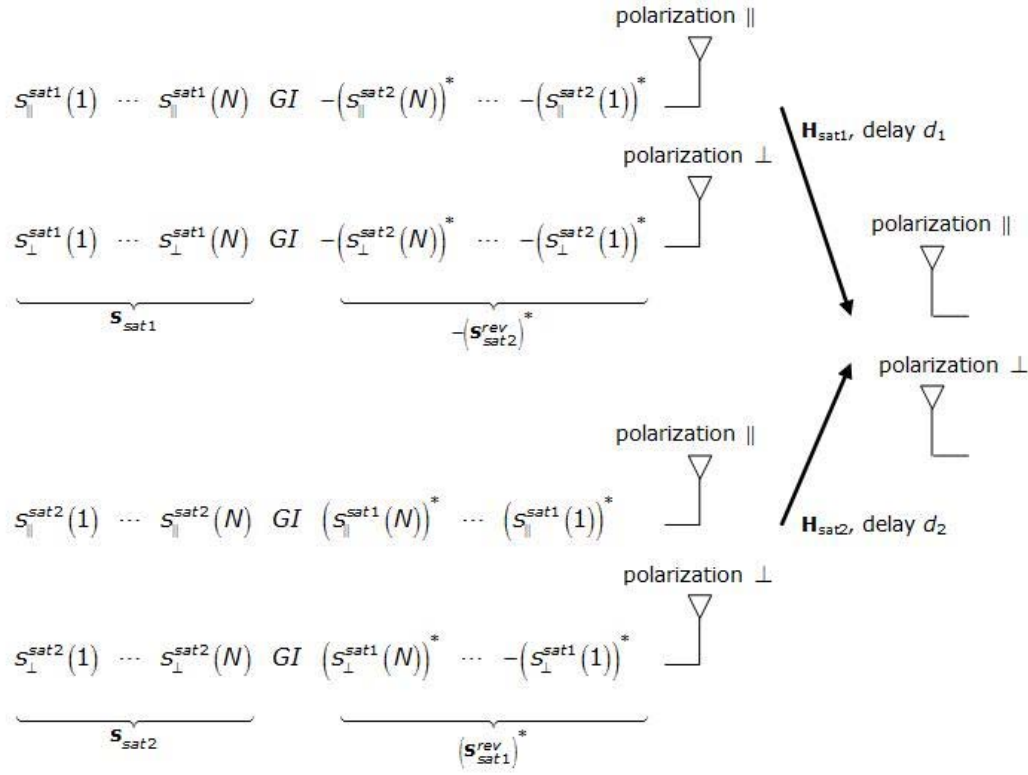


Fig. 16. Block-wise Alamouti for satellite space diversity with dual polarization [141].

I) *Improved channel modeling*: An improved model for the temporal autocorrelation of the channel is possible, profiting by the significant work carried out over the past few years by the propagation community in the area of rain attenuation time series synthesizers¹⁷.

II) *More sophisticated precoding techniques*: Linear precoding and beamforming techniques have already been the focus of [119] and [120], respectively. On the other hand, there is a growing interest on the use of nonlinear precoding over multibeam satellite scenarios, specifically THP [121], that close the gap with the theoretical DPC capacity limit.

III) *Limited feedback communications and scheduling*: Limited feedback in wireless MIMO has seen an explosive growth lately [56] due to the need of setting MIMO onto a more practical ground. Note that the introduction of any precoding strategy in commercial systems is conditioned by the required time and overhead for channel training and estimation.

IV) *Consolidation of technology*: This aims at consolidating the necessary modifications for introducing precoding into the DVB-S2 standard. The first concern is the fact that precoding, and particularly THP, excites the nonlinear behaviour of the HPA. It seems however that this is not of great concern, since multibeam payloads already tend to have HPA in multicarrier mode for flexibility reasons. The need to complement any precoding technique with a suitable predistortion technique has been acknowledged in [148], where THP is designed jointly with a waveform predistorter. A second inconsistency of MU-MIMO precoding with DVB-S2 affecting the encapsulation efficiency is that the SINR values within the same

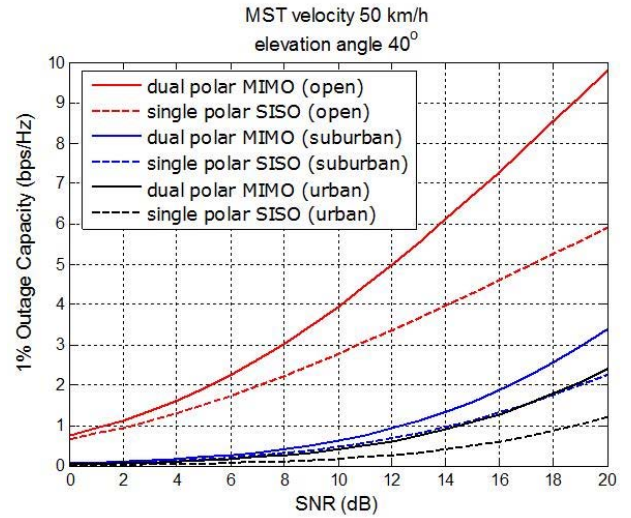


Fig. 17. 1% outage capacity of S band dual polarization MIMO MS model for open, suburban and urban area [150]. Assumed MST velocity is 50 km/h and elevation angle 40°.

beam must be similar (have a low variance) so that they are included in the same ACM format. A possible way of making precoding consistent with the ACM function of DVB-S2 is the architecture proposed and described by [125] for buffering the data prior to their processing by the DVB-S2 modulator.

B. Mobile Satellite

A lot more research efforts have been directed toward applying MIMO in MS rather than FS and the relevant

¹⁷See [66] and also the discussion in Subsection III.A.

techniques proposed seem more mature. This is especially true with regard to PTC applied to dual polarization antennas [135], [141]. A possible explanation for this may be that, contrary to FS, MS exhibits multipath due to the scatterers near the MST. Hence, MIMO techniques developed for terrestrial systems in rich multipath environments are possible candidates for MIMO MS. Moreover, current payload technology seems to be fully capable of supporting dual polarization over a single beam with full flexibility¹⁸. Possible extensions to the existing work are along the lines of the following axes:

I) *Improved channel modeling*: Since the channel effects and characteristics will ultimately determine the viability of any MIMO technique over satellite, as in FS systems, special care must be taken to furnish the most reliable channel model possible. Furthermore, although the MS channel exhibits to some extent similarities with multipath in terrestrial mobile radio, the intensity of this small scale effect is not the same, since scatterers are present only at one end of the link, and even this situation might not hold when the MSTs visit open or suburban areas. Given that only few MIMO MS channel measurements are available, the issue of a corresponding accurate channel model for MIMO MS remains open, since no proposed model can be thoroughly validated against experimental data. Nevertheless, the work along the lines of the model proposed by [137], which combines the best features from available SISO MS and terrestrial dual polarization models, is an acceptable way forward toward the end of a reliable MIMO MS channel model. Note that the single satellite/dual polarization MS channel model proposed in [137] has been extended in [150] by means of a Markov chain. These MIMO MS channel models enable a better understanding of the boundaries of MIMO performance in more realistic propagation environments. Indicatively, Fig. 17 presents results for the 1% outage capacity of a dual polarization MIMO system with respect to single polarization SISO for an MST velocity of 50 km/h and an elevation angle of 40° [150].

II) *More sophisticated space-polarization-time coding techniques*: STC for obtaining full-rate, full diversity 2x2 MIMO have recently captured the attention of wireless standardization. Among them, the most prominent ones are the Golden codes [40], briefly introduced in Section II. Hence, the application of the Golden code in a dual polarization/single satellite scenario seems as a promising near-term technique. The major concern related to Golden codes is the highly complex ML decoding necessary to profit from the whole full-rate, full-diversity frontier. A way toward reducing the complexity of ML estimation is the use of sphere decoding. Of interest are in general advanced full-rate full-diversity STC with high performance achieved with lower complexity [151], [152] since these codes naturally lend themselves to a dual polarization/single satellite scenario.

ABBREVIATIONS

3D:	Three Dimensional
3GPP:	Third Generation Partnership Project
ACM:	Adaptive Coding and Modulation
AWGN:	Additive White Gaussian Noise
BBFRAME:	Baseband Frame
BC:	Broadcasting Channel
BER:	Bit Error Ratio
BLAST:	Bell Laboratories Layered Space-Time
BS:	Base Station
CA:	Correlated Area
CCM:	Constant Coding and Modulation
CDI:	Channel Distribution Information
CDIR:	Channel Distribution Information at the Receiver
CDIT:	Channel Distribution Information at the Transmitter
CNES:	Centre National d' Etudes Spatiales
CSI:	Channel State (or Side) Information
CSIR:	Channel State (or Side) Information at the Receiver
CSIT:	Channel State (or Side) Information at the Transmitter
D-BLAST:	Diagonal-Bell Laboratories Layered Space-Time
DFE:	Decision Feedback Equalizer
DPC:	Dirty Paper Coding
DVB:	Digital Video Broadcasting
DVB-S2:	Digital Video Broadcasting-Satellite2
DVB-SH:	Digital Video Broadcasting-Satellite Handheld
EIRP:	Effective Isotropically Radiated Power
ESA:	European Space Agency
ETSI:	European Telecommunications Standards Institute
FMT:	Fade Mitigation Technique
FS:	Fixed Satellite
FST:	Fixed Satellite Terminal
GI:	Guard Interval
GSO:	Geostationary
GW:	Gateway
HAP:	High Altitude Platform
HPA:	High Power Amplifier
IDD:	Iterative Detection and Decoding
IID:	Independent and Identically Distributed
IMOB:	Improved Multibeam Opportunistic Beamforming
ISI:	Inter-Symbol Interference
ITU-R:	International Telecommunications Union-Radiocommunications
LHCP:	Left Hand Circular Polarization
LOS:	Line-of-Sight
LTE:	Long-Term Evolution
MAC:	Multiple Access Channel
MIMO:	Multiple-Input Multiple-Output
MISO:	Multiple-Input Single-Output
ML:	Maximum Likelihood
MMSE:	Minimum Mean Square Error
MOB:	Multibeam Opportunistic Beamforming
MRC:	Maximum Ratio Combining
MS:	Mobile Satellite
MST:	Mobile Satellite Terminal
MU:	Multi-User
NGSO:	Non-Geostationary
NLOS:	Non Line-of-Sight
OBO:	Output Back-Off
OD:	Orbital Diversity
OFDM:	Orthogonal Frequency Division Multiplexing
OSTBC:	Orthogonal Space-Time Block Codes
PIC:	Parallel Interference Canceler
PTC:	Polarization-Time Codes
QAM:	Quadrature Amplitude Modulation

¹⁸Benefitting, for example, from semi-active multi-matrices and flexible digital beamforming devices [149].

QoS:	Quality-of-Service
QPSK:	Quadrature Phase Shift Keying
RF:	Radio Frequency
RHCP:	Right Hand Circular Polarization
SatCom:	Satellite Communications
SatD:	Satellite Diversity
SD:	Site Diversity
SDMA:	Space Division Multiple Access
SFN:	Single Frequency Network
SIC:	Successive Interference Cancellation
SIMO:	Single-Input Multiple-Output
SINR:	Signal-to-Interference plus Noise Ratio
SISO:	Single-Input Single-Output
SNR:	Signal-to-Noise Ratio
SOF:	Start-of-Frame
STBC:	Space-Time Block Codes
STC:	Space-Time Codes
STTC:	Space-Time Trellis Codes
SU:	Single-User
SVD:	Singular Value Decomposition
TDM:	Time Division Multiplexing
THP:	Tomlinson-Harashima Precoding
UMB:	Ultra Mobile Broadband
V-BLAST:	Vertical-Bell Laboratories Layered Space-Time
XPD:	Cross-Polarization Discrimination
ZF:	Zero-Forcing
ZMCSCG:	Zero Mean Circularly Symmetric Complex Gaussian

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